Class 7

Biology

Olympiad Program
Volume - I

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1. Crop Production and Management

Solutions

LEVEL-I

1. Correct option: (A) Summer

Vegetables are frequently readily accessible in a large range during the summer. Numerous veggies may grow well in the warm climate and longer days. During the summer, there is typically an abundance of fresh, colorful vegetables at grocery shops and farmers' markets. In addition to the large selection of veggies that are offered in the summer, this is also the season when local food is at its most flavorful and fresh. Because of their nutritious value and delectable flavor, many people choose to include these seasonal vegetables in their diets.

Vegetables like tomatoes, cucumbers, bell peppers, zucchini, maize, green beans, eggplant, and a variety of leafy greens like lettuce and spinach are a few examples of those that are frequently found throughout the summer.

2. Correct option: (B) June to September.

In India, the monsoon season is known as kharif and normally lasts from June to September. Farmers plant crops at this time of year because they need the water for the crops' growth and development.

The monsoon rains' arrival provides the moisture these crops need to flourish. Rice, maize, millet, sorghum, cotton, sugarcane, and numerous vegetables are among the crops grown during the kharif season.

Crops during the Kharif season are typically sown from the start of the monsoon season to September. The collected crops can subsequently be consumed or processed further.

3. Correct option: (A) Rice

In light of its extensive global cultivation and consumption, rice has the distinction of being the world's most consumed grain. For billions of people, it is a staple dish, particularly in Asian nations where it is the foundation of the local cuisine.

Rice farming is adaptable and may be done in a variety of climates, from dry ground to flooded paddy fields.

In addition to giving farmers work opportunities and sustaining rural lifestyles, rice farming makes a considerable contribution to agricultural economies. Because of its high nutritional value, low cost, and extended shelf life, rice is an essential component of the world's food supply and is used to satisfy the dietary demands of many different people.

4. Correct option: (B) Seed drill

A seed drill is a piece of agricultural equipment made particularly for planting seeds in the ground. It is utilized for tractor-assisted sowing and makes sure that seeds are coated in soil after planting. The way the seed drill works is by making channels or furrows in the earth, and then carefully planting the seeds at the precise depth and spacing.

This exact seed-planting technique encourages consistent germination and plant development, which boosts agricultural yields. In contrast to conventional manual techniques, saves time and labor while preventing birds from eating the seeds. The seed drill also facilitates appropriate seed-to-soil contact for efficient germination, optimizes seed dispersion, and reduces seed loss.

5. Correct option: (C) Srivari

The Srivari kind of paddy is the one that needs a bigger space between plants than the other varieties. This type is renowned for its robust growth and wide-ranging canopy, which need greater room between plants for the best growth.

Srivari paddy plants may obtain enough sunshine, air circulation, and nutrients by having enough space between them, which results in greater yield and healthier plants. To satisfy its growth requirements, farmers that cultivate Srivari paddy should maintain a greater space between plants.

6. Correct option: (A) Tikka disease

Tikka disease is a typical disease that affects groundnuts. The fungus Cercospora arachidicola is what causes tikka illness. Small, dark brown to black spots with a concentric ring pattern appear on the groundnut plant's leaves and pods as a result of this condition. The spots may get larger and combine as the illness spreads, which can cause defoliation, yellowing of the leaves and pods, and decreased production.

Tikka disease in groundnut agriculture may be controlled and its effects reduced with the use of effective management techniques such crop rotation, seed treatment, and foliar fungicide use.

7. Correct option: (D) Irrigation

Irrigation is the systematic, planned application of water to plants in order to promote their optimum development. In order to maintain soil moisture levels, it entails the controlled administration of water to crops at predetermined intervals. Since it guarantees that plants receive enough water for photosynthesis, nutrient absorption, and general growth, adequate irrigation is essential for crop health. The kind of crop, soil, weather, and development stage are just a few examples of the variables that affect when and how often irrigation should be applied.

Different irrigation techniques are used to effectively distribute water to crops, minimising water loss via evaporation and guaranteeing targeted delivery to plant roots. These techniques include sprinkler systems and drip irrigation systems. These techniques support sustainable agricultural practises and water resource conservation.

8. Correct option: (A) Dried

Grain must be adequately dried in order to lower its moisture content before being stored. Harvested grains frequently have significant moisture content, which, if left untreated, can cause deterioration and the development of microbes.

Using the sun to dry grains is a popular and efficient way to get rid of extra moisture. The grains are spread out in thin layers and exposed to sunshine to dry by evaporating the moisture. The quality and integrity of the grains are protected during storage by adequate drying, which inhibits the growth of bacteria, fungus, and insect pests that flourish in damp settings.

The acceptability of grains for food and retaining their viability for germination if used as seeds is ensured by well dried grains, which also have a lower risk of mould development, mycotoxin contamination, and insect infestation. Additionally, before storage, grains can be dried to increase their shelf life and maintain their nutritious content.

9. Correct option: (A) Broadcasting.

Broadcasting is the method of seed dispersal by sprinkling or scattering seeds evenly over a wide area. In agriculture, broadcasting is commonly used for sowing seeds in fields, especially for crops like grass, grains, and certain vegetables. The seeds are dispersed by hand or by using seeders, ensuring they are spread uniformly across the soil surface.

This method allows for a relatively even distribution of seeds, promoting uniform plant growth and reducing competition among individual plants for resources like water, sunlight,

and nutrients. Options B) Throwing and C) Flying do not accurately describe the method of seed dispersal, and option D) None of these is not the correct answer in this context.

10. Correct option: (C) Cold storage

Agricultural products may often be kept for a long time by being placed in cold storage. It entails keeping perishable goods at low temperatures, including various cereals, dairy products, and produce. The purpose of cold storage facilities is to maintain particular temperature and humidity levels that inhibit the growth of bacteria and enzymes that cause rotting and degradation. By maintaining low temperatures, typically ranging from 0 to 10 degrees Celsius (32^0 to 50^0 Fahrenheit), cold storage inhibits the activity of spoilage-causing agents and extends the freshness of the stored products.

Agricultural goods can stay fresh, maintain their nutritional content, and have a longer shelf life when they are kept in cold storage since the rate of deterioration is greatly lowered. Options A) Bamboo bin and B) Metallic bin are not specifically associated with preserving agricultural products for longer periods, and option D) None of these is not the correct answer in this context.

11. Correct option: (B) Sank seeds

Seeds that sink in water generally have a higher likelihood of germinating well due to several key factors. Firstly, sinking seeds indicate higher seed viability, suggesting that they are alive and capable of germination. Secondly, sank seeds often have intact seed coats, providing protection and allowing water to penetrate for germination. Additionally, these seeds typically possess well-developed embryos, containing essential structures for growth. Sank seeds also tend to have adequate internal reserves of nutrients and energy, providing nourishment during early stages of germination.

Moreover, sinking seeds are less likely to have encountered damage or decay, reducing the risk of unsuccessful germination. While the sinking or floating behavior is an approximate indicator, other factors such as seed age, genetics, and specific environmental requirements also influence germination success.

12. Correct option: (A) Rabi

Wheat is primarily cultivated during the Rabi season due to its favorable climatic conditions and the crop's growth requirements. First of all, because wheat is a cool-season crop that prefers moderate temperatures for growth, the Rabi season's milder weather is suitable.

Second, the availability of water is very important. During the Rabi season, the soil moisture left over from the previous monsoon season is often sufficient, decreasing the need for irrigation.

Thirdly, the Rabi season's shorter days and longer nights give wheat the proper photoperiod for its reproductive development stage. In addition, the number of pests and illnesses that generally harm wheat decreases during the Rabi season, lowering the possibility of crop loss. Finally, the Rabi season harvest coincides with the arrival of spring, ensuring an adequate supply of wheat for consumption and distribution.

13. Correct option: (A) Rabi

The term "Rabi crops" refers to plants that are only cultivated during the winter. Crops during the rabi season are planted in the months of October to December and harvested in the following spring. These crops include, among others, wheat, barley, mustard, peas, gramme, and linseed. Since these crops are grown in the winter and harvested in the spring, the word "Rabi" is derived from the Arabic word for "spring." The milder temperatures, prolonged photoperiods, and lingering monsoon-season soil moisture are beneficial to rabi crops. They are essential in maintaining the availability of food and giving farmers a reliable source of income during the winter.

14. Correct option: (B) Kharif

Kharif crops are those that are raised during the rainy season. Crops known as kharif are planted at the start of the monsoon season in June or July, and they are harvested from September to October. These crops include, among others, groundnuts, cotton, rice, maize, millet, and sorghum. Given that they need more water for growth and development, kharif crops thrive in the hot, muggy weather of the rainy season. They usually have a lengthier growth cycle compared to Rabi crops and mainly rely on the monsoon rains for irrigation. In areas with a distinct monsoon pattern, kharif crops are essential for agriculture because they make a substantial contribution to total food supply and the rural economy.

15. Correct option: (D) All the above

Urea, DAP (diammonium phosphate), and potash are a few typical chemical fertilizers used in agriculture. The fertilizer urea, which is based on nitrogen, gives plants the nitrogen they need for leaf formation and general growth. The root growth, blooming, and fruiting of plants are supported by the phosphorus-based fertilizer known as DAP. The term "potash," on the other hand, refers to fertilizers that contain potassium, a mineral that is crucial for plant health, disease resistance, and a number of physiological functions. These chemical fertilizers are essential in refilling the soil's nutrient content, enhancing plant nutrition, and increasing crop yields. To reduce any possible negative effects on the environment, it is crucial to utilize them wisely and in the right amounts.

16. Correct option: (C) V

V-shaped ridges are indeed formed in certain farming practices when ploughing is done. This technique is known as "ridge and furrow" or "ridge tillage." It involves creating raised V-shaped ridges of soil with furrows in between. The purpose of this method is to improve water management and drainage in areas with high rainfall or poor soil drainage. The V-shaped ridges help divert water away from the planting area, preventing waterlogging and promoting better root aeration. The furrows between the ridges collect excess water and serve as channels for irrigation or drainage. This technique is commonly used in certain crops such as potatoes, where it helps in improved root development and better access to sunlight and nutrients.

17. Correct option: (B) 2300 BC

The Harappan civilization, thriving around 2300 BC in the Indus River valley, displayed remarkable advancements in crop management and cultivation. Agriculture played a crucial role in their economy, and they cultivated a variety of crops. The Harappans employed sophisticated techniques for irrigation, utilizing an extensive network of canals and wells to ensure a steady water supply for their fields.

The irrigation systems supported year-round cultivation, enabling multiple cropping seasons and maximizing agricultural productivity. They practiced systematic land preparation, including ploughing and leveling, to create fertile fields. The Harappans also employed innovative strategies for water management, such as the construction of reservoirs and the implementation of water storage and distribution systems. Evidence suggests that they practiced crop rotation to maintain soil fertility and prevent nutrient depletion.

Additionally, the Harappans may have utilized organic manure and composting methods for soil enrichment. Their agricultural practices contributed to the sustenance and prosperity of the civilization, ensuring a stable food supply for its population.

18. Correct option: (C)

19. Correct option: (A) 63^0

In pasteurization, milk is heated at a temperature of approximately 63^0 Celsius (145 degrees Fahrenheit). This specific temperature is chosen as it effectively kills harmful bacteria and

pathogens present in the milk, reducing the risk of foodborne illnesses. The heat treatment destroys most of the pathogenic microorganisms while still preserving the nutritional properties and sensory characteristics of the milk.
By subjecting milk to this controlled heating process, pasteurization ensures that consumers can enjoy a safer product without compromising the taste and quality. Pasteurized milk is widely consumed and trusted as a safe source of nutrition.

20. Correct option: (D) Chilka

The Chilka buffalo is the traditional breed of buffalo used for cattle in Odisha. In particular, this breed has been suited to the marshes and marshy terrain of the Chilka Lake region of Odisha. Chilka buffaloes are prized for their prowess in this environment, which makes them suitable for grazing and draught uses. They are also well recognised for producing milk, which helps the local dairy sector. The Chilka buffalo is significant culturally and economically in Odisha, where it is crucial to local populations' lives and agricultural practises.

21. Correct option: (B) Eggs

Layers refer to a specific type of poultry breed that is primarily raised for egg production. These birds are selectively bred to lay a large number of eggs throughout their productive life. They are known for their high egg-laying capacity and can produce a significant quantity of eggs on a regular basis. Layer chickens are typically housed in specialized facilities with proper lighting, temperature control, and nutrition to ensure optimal egg production. The eggs produced by layers are a valuable source of protein and are widely consumed as a nutritious food item.

22. Correct option: (D) January to April

The months of January to April generally see higher egg prices. During this period, there are various factors that contribute to the increased prices, such as increased demand due to seasonal festivities like Easter and the onset of spring when people tend to consume more eggs. Additionally, during these months, there may be fluctuations in the supply of eggs due to factors like weather conditions, changes in feed availability, and production cycles. These factors collectively contribute to the higher prices of eggs during the months of January to April.

23. Correct option: (A) Aseel

Aseel, also known as Asil or Asli, is a breed of chicken that is well-known and valued for its fighting abilities. Aseel chickens have a long history in India and are highly regarded for their strength, endurance, and aggressive nature. They are bred specifically for cockfighting, a traditional sport in which two Aseel roosters are pitted against each other in a fight. Aseel chickens are characterized by their muscular build, strong beaks, and distinctive appearance. They are considered a prized breed among enthusiasts of cockfighting and are highly sought after for their fighting prowess.

24. Correct option: (A) 40kg

Emus are the second-largest living birds by height and are native to Australia. These flightless birds have a large and sturdy build, and adult Emus typically weigh around 45 to 60 kg (99 to 132 pounds). However, there can be variations in the weight of individual birds based on factors such as age, gender, and overall health. The weight of an Emu bird can also be influenced by environmental factors and the availability of food resources.

25. Correct option: (D) 1

In a honey bee colony, there is typically only one queen bee. The queen bee is the sole reproductive female in the colony and plays a crucial role in the survival and functioning of the colony. She is larger in size and specially developed for egg-laying. Her primary responsibility is to lay fertilized eggs, which will develop into worker bees and new queens.

The queen bee releases pheromones that maintain harmony and coordination within the colony. The presence of multiple queen bees in a single colony can lead to conflicts and disruptions in the social hierarchy. Therefore, in a healthy and stable honey bee colony, there is generally one queen bee.

26. Correct option: (B) Agriculture

Agriculture encompasses various activities related to the cultivation of crops and the rearing of animals. Animal husbandry specifically focuses on the management and care of domesticated animals for agricultural purposes. It involves practices such as breeding, feeding, housing, and healthcare of livestock, including cattle, poultry, sheep, goats, and pigs. Animal husbandry plays a vital role in agriculture as it provides a sustainable source of food, fiber, and other products derived from animals. It contributes to the overall agricultural economy and helps meet the growing demand for animal-based products such as meat, milk, eggs, and wool. Additionally, animal husbandry practices are crucial for maintaining the health and productivity of livestock, ensuring optimal production and maximizing agricultural outputs.

27. Correct option: (D) Nagori

The Nagori is a traditional variety of cow found in certain regions of India. The regions of Rajasthan, India, known as Bikaner, Jodhpur, and Nagaur, are home to the Nagori breed of cattle. With a long, thin face like a horse's, these animals are renowned for their upright, nimble character. Due of their close closeness to the Hariana and Kankrej breeds, it is probable that these breeds have had an impact on them.

The bodies of nagori cattle are excellent, massive, and deep, with straight backs and well-developed quarters. They are known for their trotting draught prowess and are frequently utilised in light iron-wheeled carts for swift transportation in the Rajputana region of Rajasthan.

These cows produce 603 kg of milk on average every lactation, with a milk fat level of 5.8% on average. Nagori cattle are valued for their distinctive qualities and cultural significance in the local populations. They also support local agricultural practises.

28. Correct option: (D) Both A and B

D) Both A) Jersey and B) Holstein.

The Jersey breed is known for its milk production capabilities, yielding an average of 5,000-8,000 kg of milk per lactation. A Jersey cow can produce around 20 liters of milk per day, making it a high-yielding breed. Crossbred Jersey cows also have good milk production, typically giving 8-10 liters per day. Jerseys are well-suited to hot and humid climates, making them suitable for the climatic conditions of India.

On the other hand, the Holstein breed is considered one of the best dairy breeds among exotic cattle in terms of milk yield. They have an average milk yield of 7,200-9,000 kg per lactation and can provide around 25 liters of milk per day. Crossbred Holstein cows can produce 10-15 liters of milk per day. Holsteins are large, capacious animals and are adaptable to coastal and delta areas.

29. Correct option: (A) 63^{0}C for 30 mnts

During pasteurization, milk is heated to a temperature of approximately 63°C (145°F) and maintained at that temperature for about 30 minutes. This specific temperature and duration are carefully chosen to effectively destroy harmful bacteria and pathogens present in the milk, ensuring its safety for consumption. The heat treatment kills potential pathogens such as Salmonella, Escherichia coli (E. coli), and Listeria, reducing the risk of foodborne diseases.

The pasteurization process also helps extend the shelf life of milk by inhibiting the growth of spoilage-causing microorganisms. By subjecting milk to this controlled heat treatment,

pasteurization strikes a balance between eliminating harmful bacteria while retaining the nutritional qualities and sensory attributes of the milk.

30. Correct option: (A) Food grains

From the middle of the 1960s onward, India saw a period of agricultural transformation known as the "Green Revolution." It required implementing cutting-edge agricultural techniques and tools, including the use of High Yielding Variety (HYV) seeds, better irrigation techniques, herbicides, and fertilisers. M.S. Swaminathan, known as the "Father of the Green Revolution" in India, was a key figure in advancing these ideas.

In areas like Punjab, Haryana, and Uttar Pradesh, the Green Revolution led to a huge rise in food grain output, particularly for grains like wheat. Modern agricultural techniques and the introduction of HYV seeds industrialised Indian agriculture, increasing output and alleviating food shortages.

31. Correct option: (A) 3rd

India is the third-largest egg producer in the world. The poultry business in the nation has expanded significantly, which has led to a huge rise in egg output. The Indian poultry industry has been effective in supplying eggs to fulfil the expanding local and international demand. India has become a significant producer of eggs, contributing significantly to global egg production because to improvements in chicken farming techniques, enhanced genetics, and effective production techniques.

32. Correct option: (C) 6th

India is the world's sixth-largest producer of chicken. With rising demand for chicken meat as a source of protein, India's poultry sector has seen significant expansion. To fulfil the expanding demands of customers, the nation has achieved considerable advancements in breed improvement, production methods, and commercial poultry farming practises. In addition to meeting domestic and international demand for chicken meat, India's poultry industry is essential for creating jobs and boosting the nation's economy.

33. Correct option: (C) Meat and eggs

Breeds of chicken that are often raised for both meat and eggs include Rhode Island Red and White Plymouth. These breeds are well-known for having traits that make them good for both producing meat and eggs. They may lay a sizable amount of eggs and have high-quality meat.

Due to its capacity to provide a sizable number of both meat and eggs, the White Plymouth Rock breed, also known as White Rocks, is a preferred option for commercial poultry production. On the other hand, Rhode Island Reds are renowned for having good meat yields along with exceptional egg-laying ability. Overall, these breeds are favoured by farmers who seek to maximise the amount of meat and eggs their chickens produce.

34. Correct option: (A) 21 days

Prior to hatching, hens normally give their eggs a 21-day incubation period. The hen supplies the essential warmth and humidity during this time so that the eggs can effectively grow and hatch. The majority of chicken breeds naturally incubate their eggs for 21 days. In order to guarantee appropriate embryo growth during this period, it is crucial to maintain the eggs at a constant temperature. The eggs will hatch and the chicks will break out of their shells after 21 days. The continuance of the chicken species depends on this period, which is a crucial component of the reproductive cycle.

LEVEL-II

1. Correct option: (C) Management of plants and animals

Animal and plant management are within the purview of agricultural science. It includes a diverse variety of academic fields and professional techniques designed to raise agricultural production and sustainability. Understanding and using knowledge of plant biology, crop production, soil science, animal husbandry, and genetics are required in this discipline.

The goal of agricultural science and practise is to improve the care and management of animals in agricultural systems as well as the development, health, and productivity of plants, such as crops. The science of agriculture aims to enhance food production, environmental preservation, and the general wellbeing of agricultural systems and societies by combining the management of both plants and animals.

2. Correct option: (A) Mentha

The herb mentha, often known as mint, is not generally categorised as a weed. The fragrant leaves of mint plants are frequently grown for use in food preparation and medicine. Mint is often not regarded as a weed species, despite the fact that it has the potential to spread and become invasive in some situations.

On the other hand, weed species including amaranthus, chenopodium, and wild oat are well known. A grassy plant called wild oat (Avena fatua) can lower productivity by competing with crops. A group of plants known as Chenopodium include a number of common weed species, including lamb's quarters (Chenopodium album). The plant genus Amaranthus, sometimes referred to as pigweed, has a number of troublesome species in agricultural fields.

3. Correct option: (C) By practicing crop rotation

Crop rotation can be used to restore soil fertility. Crop rotation is the methodical, deliberate alternating of several crops in a set order throughout time. Every crop has varied nutritional needs and engages with the soil in various ways. By changing up the crops that are grown, soil fertility is increased and nutrient depletion is prevented. Different crops also improve soil health by retaining or supplying certain nutrients, disrupting the life cycles of pests and diseases, lowering weed pressure, and improving soil structure. Crop rotation is a practical and efficient way to manage and replace soil fertility in agricultural systems.

4. Correct option: (B) Aluminium sulphate

Aluminium sulphate is not regarded as a chemical fertiliser. Instead of serving as a source of nutrients for plant development, it is largely utilised to modify soil and treat water. In several industrial and agricultural applications, aluminium sulphate is used to reduce the pH of water and to modify the pH of soil in alkaline soils. On the other hand, chemical fertilisers include urea, potash, and NPK.

Fertilisers that are based on nitrogen include urea, potash, and NPK, which is a complex fertiliser made up of nitrogen, phosphorus, and potassium. These chemical fertilisers are especially designed to provide vital nutrients that support plant growth, development, and productivity.

5. Correct option: (A) Cu, Zn, B

Micronutrients are necessary substances that plants must have in relatively little amounts, yet they are crucial to many physiological functions. Micronutrients include elements such as boron (B), copper (Cu), and zinc (Zn). These micronutrients are essential for the development, growth, and general health of plants. Photosynthesis and enzyme activation both include copper. Protein synthesis, hormone control, and enzyme activity all depend on zinc. The synthesis of cell walls, pollen germination, and nutrient absorption are all influenced by boron. These micronutrients are crucial for healthy plant development and are frequently added to the soil or fertiliser for plants when shortages are noticed.

6. Correct option: (C)

When harvesting and threshing crops like wheat, corn and soybeans, a combine, often referred to as a combine harvester, is employed. The crop harvesting process is streamlined by combining the duties of harvesting and threshing into a single operation. To collect mature crops like wheat stalks or maize plants, the combine utilises a cutting mechanism. The threshing process separates the grain (such as wheat or maize kernels) from the remaining plant matter (stalks, husks, etc.) as the harvested crop enters the machine.

In the combine, the separated grain is gathered in a storage tank and made ready for processing or transportation. The remainder of the plant is ejected or thrown away. By drastically lowering labour, combines have revolutionised agriculture.

7. Correct option: (D) Sprinkler system

Rotating nozzles are utilised in the Sprinkler System while discussing irrigation techniques. Sprinkler systems are intended to disperse water across fields similarly to rainfall. By spraying water in a revolving pattern, the spinning nozzles are essential in providing uniform water coverage.

The sprinkler heads revolve to ensure that water is uniformly distributed around the field, boosting effective water use and minimising water waste. This technique is frequently used in agriculture for extensive irrigation since it enables efficient watering of a sizable region while upholding water conservation and maximising crop water needs.

8. Correct option: (A) Traditional method of irrigation

Moats are tiny holes or pits that are excavated to plant seeds or seedlings; this has less to do with irrigation and more to do with seed sowing or planting methods.

Instead of irrigation, dhekli is a technique used to flatten or crush soil clods following ploughing, which is a part of land preparation.

Rahat is a method of soil preparation that also entails turning over and loosening the soil with a tool like a plough.

Moat, Dhekli, and Rahat do not fall under the conventional technique of irrigation if we just take option A) into account.

9. Correct option: (B) Beans

Starchy grains that are predominantly utilised as food sources are produced by plants known as cereal crops. These grains are significant staple crops globally and are frequently high in carbs. Wheat, millet, and barley are a few examples of cereal crops.

Conversely, beans are a member of the family of legumes. Various plants that produce pods with seeds within are referred to be legumes. Although they are valued and very nutritious, beans are not considered a cereal crop. Beans, on the other hand, belong to the legume family and are categorized as pulses or legume crops. They have a greater protein content than cereals and are renowned for their capacity to fix nitrogen in the soil, enhancing soil fertility.

10. Correct option: (C) Dried neem leaves

Since dried neem leaves naturally resist insects, they are utilised as grain storage at home. Neem leaves contain substances like azadirachtin, which are powerful insect repellents. Neem leaves are completely dried to improve their preservation qualities before being used to store grains. After that, they are positioned or dispersed among the grains that have been kept to form an insect and pest-resistant barrier.

The dried neem leaves' strong perfume serves as a natural insect repellent, keeping pests away from the grains. Using dried neem leaves in this manner is a traditional strategy to safeguard grains that are kept in storage, assuring their long-term preservation and averting any harm from pests or insects.

11. Correct option: (C) Cattle dung and crop residue

Farmyard manure (FYM), sometimes referred to as organic manure or FYM, is a form of natural fertiliser and soil amendment that is produced by the decomposition of a combination of agricultural debris and animal dung. A high amount of organic matter and nutrients is available in cattle dung, which is made up of the faeces and urine of cattle.

The organic component of the manure also benefits from agricultural waste, such as residual plant matter from harvested crops. These materials breakdown naturally, such as by composting, generating a nutrient-rich mixture that may be added to soil to increase its fertility, structure, and general health.

12. Correct option: (B) Roots of leguminous plants

The majority of rhizobium bacteria are found in nodules that form on the roots of leguminous plants. Rhizobium bacteria are recognised for their capacity to fix atmospheric nitrogen. The Rhizobium bacteria and legume plants have a special symbiotic connection. The bacteria populate the root nodules and transform air nitrogen into a form the plant can use.

Leguminous plants can obtain a crucial ingredient for their growth and development through a process known as nitrogen fixation. Leguminous plants may grow in nutrient-poor soils thanks to Rhizobium, which is found in the root nodules. Rhizobium also plays a critical role in enhancing soil fertility by converting atmospheric nitrogen into a form that can be used by plants.

13. Correct option: (D) Ganga 5

Among the available possibilities, Ganga 5 is the hybrid variety of maize. Different parent lines are crossed to create hybrid types in order to produce offspring with the desired qualities. A hybrid maize variety with certain properties such a higher yield potential, improved disease resistance, or other beneficial attributes has been carefully cultivated, and this trait is what is meant by the name Ganga 5. Due to its enhanced characteristics, this specific hybrid variety has become well-known and well-liked among farmers and is frequently used in agricultural practises.

14. Correct option: (A) Plough, hoe cultivaor

The plough, hoe, and cultivator are the primary implements used to break the soil into the correct grain size and provide a greater yield before sowing the seeds.

A instrument used to turn over and loosen the soil is the plough. It assists in loosening up compacted soil, burying agricultural waste, and improving soil drainage and aeration.

A portable device called a hoe has a flat, sharpened blade. It is used to break up soil clumps, get rid of weeds, and provide a smooth, level seedbed.

The cultivator is a device with several blades or tines. It is used to cultivate the soil only superficially, break up clumps, and incorporate organic materials or fertilisers into the soil's topmost layers.

15. Correct option: (D) Seeds placed deep into soil

Seeds are typically planted at a suitable depth, which varies depending on the type of seed and plant. Planting seeds at an appropriate depth is crucial for their successful germination and growth. When seeds are buried too deep in the soil, they may struggle to reach the surface and access sunlight, hindering their ability to undergo photosynthesis.

Additionally, planting seeds too deep can limit their access to moisture, impeding their ability to absorb the necessary water for germination. Furthermore, an inadequate exchange of gases between the seed and its surroundings may occur, depriving the seed of the oxygen it needs for metabolic processes. Therefore, it is important to follow the recommended planting depth for each type of seed to provide them with the best conditions for sprouting, establishing roots, and developing into healthy plants.

16. Correct option: (C) Drip system

The drip irrigation system is not a common type of irrigation. It is a contemporary irrigation technology that uses a system of tubes or emitters to precisely and carefully apply water to the plant's root zone. Drip irrigation is a prominent option in contemporary agricultural practises because of its reputation for water efficiency and precise delivery.

On the other hand, conventional irrigation techniques include the chain pump, pulley system, and lever system. These techniques, which have been used for millennia, employ mechanical or human means to move water for irrigation of crops.

17. Correct option: (D) Rhizome

A rhizome, often known as an underground modified stem, is an example of ginger. There are nodes and internodes on rhizomes, which are horizontal subterranean stems. The rhizome of ginger acts as a nutrient storage organ and promotes vegetative growth. The ginger plant may extend and expand as a result of the new branches and roots it generates. Ginger's rhizome is an important and well-known example of an underground modified stem since it is picked for both culinary and medicinal purposes.

18. Correct option: (D) None of the given options

A) Plants benefit from the growth of living organisms like bacteria and earthworms. It's accurate to say this. When it comes to the health of the soil and the cycling of nutrients, earthworms and microbes are key players.

B) The live component is supported by non-living elements. This assertion is true. Living things, such as plants and microbes, depend on non-living elements like soil, air, water, and minerals for their physical and chemical sustenance.

C) The seed drill saves labour and time. This assertion is true. In contrast to conventional hand sowing techniques, a seed drill is a mechanical tool used for accurate and effective seed planting.

Since all the given statements are correct, the correct answer is D) None of the given options.

19. Correct option: (C) Neem leaves

Neem leaves have been used for centuries as a natural pesticide. They include substances that serve as antifeedants, insect repellents, and growth and reproduction disruptors. Crushed neem leaves, neem oil extracts, or formulations containing neem can all be used to protect crops from insects and pests. Neem leaves also have herbicidal qualities that inhibit the growth of weeds in agricultural regions.

Neem leaves are well known for their strong pesticidal powers and are frequently employed by farmers to safeguard their crops against weeds and insect pests. Other leaves, such as basil leaves, peepal leaves, and mango leaves, may have diverse useful features.

20. Correct option: (A)

The International Rice Research Institute (IRRI), which is situated in the Philippines, brought a variety of rice known as the IR variety to India. The IR rice variety is well known for its qualities of disease and insect resistance, which have made it an important contribution to rice agriculture in several nations, including India. The introduction of the IR variety has improved crop resilience, increased rice yield, and aided with efforts to ensure food security.

This sentence makes it clear that the IR variety is only related to rice and that it was brought to India from the Philippines. Rice production has benefited greatly from the IR variety, which has given farmers access to enhanced crop types that are resistant to a wide range of diseases and pests. This rice variety was introduced to help farmers in India and other nations overcome agricultural obstacles and improve the overall productivity and calibre of rice crops.

21. Correct option: (C) Sharbati, Sonara

It is well known that Dr. M.S. Swaminathan made significant advances to wheat farming during India's Green Revolution. His creation and promotion of high-yielding dwarf wheat cultivars like "Kalyan Sona" and "Sonalika" revolutionised the industry, boosting yields and solving issues of food security. His work had a tremendous influence on Indian agriculture, enhancing wheat farming and assisting in ensuring that the populace had access to food.

Even though "Sharbati Sonara" isn't particularly linked to Dr. M.S. Swaminathan, it is a well-liked wheat variety in some parts of India that is renowned for having a high protein content. However, it's critical to recognise how Dr. Swaminathan's introduction of high-yielding dwarf cultivars has had a larger influence on wheat farming in India.

22. Correct option: (B) It has a balance of all plant nutrients

Compost or farmyard manure are examples of organic manures that come from natural sources like plant remnants, animal dung, or organic wastes. Although organic manure is good for the health and fertility of the soil, it can not always contain a balance of all the plant nutrients.

Nitrogen, phosphorus, and potassium are among the many elements that organic manure normally includes, although the amount of each component varies depending on the source and makeup of the manure. Comparatively to synthetic fertilisers, which are especially designed to offer a balanced supply of vital plant nutrients, organic manure often has a lower nutritional concentration.

23. Correct option: (B) Leguminous plant can fix nitrogen

Unique to legume plants, rhizobia, which fix nitrogen, create a symbiotic connection with legume plants. These microorganisms are found in nodules in the roots of leguminous plants, where they transform atmospheric nitrogen into a form the plants can use for growth. Leguminous crops may naturally replenish the soil with nitrogen through a process known as nitrogen fixation, which is one of its major advantages.

Option a) Chain pumps are not the standard irrigation technique. Traditional irrigation systems use methods such employing canals, wells, or water channels.

Option C) Soil nutrients are not always depleted when various crops are grown in different seasons on the same field. In reality, by limiting the excessive depletion of certain minerals and lowering the danger of pests and diseases, crop rotation and diverse cropping systems can assist preserve soil fertility.

24. Correct option: (C)

The process of planting two or more crops at once on the same plot of land is known as intercropping. It entails planting various crops next to one another in a planned sequence or arrangement. The crops used for intercropping may have advantageous interactions, complimentary growth patterns, or nutritional requirements. By diversifying crops and encouraging optimal use of sunshine, water, and nutrients, intercropping helps maximise land production, optimise resource usage, and lower the chance of crop failure.

25. Correct option: (A) Give rise to sexually sterile plants

The 'Terminator Seed Technology,' often referred to as Genetic Use Restriction Technology (GURT), is a contentious method of genetically altering plants to generate sterile or incapable of germination seeds. This suggests that option A, sexually sterile plants, should grow from the seeds generated by the method.

Farmers are compelled to buy new seeds for each planting season due to the terminator seed's ability to prohibit farmers from conserving and recycling seeds from previous crops. Farmers are extremely concerned about this element of terminator seeds since it limits their customary practise of seed storage and traps them in a cycle of dependency on seed providers.

26. Correct option: (D)

27. Correct option: (C) Only 1 and 4

Vitamin A: A lot of vitamin A may be found in fish liver oil. The immune system, promotion of healthy skin, and maintenance of excellent vision all depend on vitamin A.

Vitamin D: Vitamin D is also abundant in fish liver oil. Vitamin D is essential for boosting calcium and phosphorus absorption, supporting strong bones and teeth, and sustaining general bone health.

28. Correct option: (A)1,2 & 3

Introduction: To improve a crop population's qualities, additional genetic material or varieties are bred into the population. This may entail bringing in fresh plant kinds from other areas or adding genetic material using strategies like genetic engineering.

Hybridization: The process of breeding wherein two genetically dissimilar plant species are crossed to produce a hybrid progeny. Hybridization can produce plants with desired features like higher production, disease resistance, or enhanced quality.

Selection: Selecting and carefully breeding plants from a population of plants that have qualities you want. Plants that exhibit qualities like increased yield, better quality, or resistance to pests and diseases are chosen and allowed to breed, which over several generations improves the crop population.

Segregation: Refers to the separation of characteristics rather than active improvement, is not required for raising crop quality.

The techniques required to increase crop quality include introduction (introducing new varieties), hybridization (crossing various varieties), and selection (selecting and propagating favourable features). As a result, the right response is A) 1, 2 & 3 (Introduction, Hybridization, and Selection).

29. Correct option: (C) Only 1, 2 and 3

Paddy (Rice): Growing paddy or rice frequently uses the transplanting method. With this technique, rice seedlings are first raised in a separate nursery field and then, once they reach a specific development stage, moved to the main paddy field. In addition to ensuring ideal growth conditions for the rice crop, this enables improved control over plant spacing.

Tomato: Growing tomato plants is frequently accomplished by transplantation. First, tomato seeds are sown inside or in a greenhouse in seed trays or pots. The seedlings are moved into the outdoor garden or field once they have grown to an appropriate size and strength. This technique, especially in areas with shorter growing seasons, enables early germination and gives the plants a head start.

Chilli (Chili peppers): Similar to tomato plants, chilli peppers are frequently cultivated using the transplanting technique. The seedlings can develop and flourish in the best conditions since the seeds are originally planted in controlled surroundings like seed trays or pots. The chilli pepper seedlings are moved into the actual growing area, such as a garden or field, once they reach a particular size.

Millet: Millet crops are typically not grown using the transplantation method. Instead, millet seeds are usually directly sown into the ground or field where they will grow.

30. Correct option: (C) a – ii; b – i; c – iii

a) Sowing: The act of dispersing seeds for growth is referred to as sowing. The first stage of planting crops is this. Option ii correctly identifies this agricultural practise as "sowing," which is the act of dispersing seeds on the earth.

b) Irrigation: Water is supplied to crops in the fields during irrigation. It is crucial for giving plants the moisture they need for development and growth. This practise is adequately described by option i, which explains that irrigation is the process of providing water to crops in the fields.

c) Weeding: The act of eliminating undesirable plants from a field is referred to as weeding. It entails getting rid of undesired plants or limiting their development so they don't compete with cultivated crops for resources like water, nutrients, and sunshine. Weeding is correctly described as the process of eliminating undesirable plants from a field in Option iii.

2. Adaptation of Animals to Climate

Solutions

LEVEL-I

1. Correct option: (A) Very hot and wet

The climate found in Kerala is categorized as "very hot and wet". Accordingly, Kerala has high temperatures and humidity all year round. The area has high rainfall throughout the monsoon season, which adds to its profusion of wildlife and beautiful foliage. The distinctive natural beauty and picturesque landscapes of Kerala are mostly due to its warm and humid climate, which also makes it a great location for a variety of plant and animal species to thrive.

2. Correct option: (A) In the afternoon

When the sun rises, the temperature reaches its peak for the day. The Earth's surface starts to warm when the sun rises, which raises temperatures throughout the morning and early afternoon.

Although the maximum temperature is normally measured around midday when the sun is at its highest point in the sky, this heating impact peaks in the late afternoon. After then, when the sun sets and the Earth radiates heat away, the temperatures begin to steadily fall. Thus, the hottest time of the day often occurs in the morning, around noon, before it begins to drop off in the evening.

Early in the day, when the sun is low in the sky and its rays are less direct, the temperature is at its lowest of the day. The air above the Earth's surface has cooled down throughout the course of the previous night.

3. Correct option: (B) Sand cools down faster

Nights in deserts are much cooler primarily because of the low heat-retaining capacity of sand. During the day, the desert sand absorbs a significant amount of heat from the sun and becomes very hot. However, as the sun sets and the desert loses its direct exposure to sunlight, the sand cools down rapidly due to its low heat retention properties.

This leads to a substantial drop in temperature, making desert nights considerably cooler compared to the scorching daytime temperatures. Option (b) correctly identifies this phenomenon as the reason for cooler nights in deserts.

The other choices are incorrect. In deserts, the humidity does not rise significantly at night. Although wind does blow in deserts, it does not blow hard enough to considerably chill the air. Deserts have no nighttime precipitation.

4. Correct option: (A) Climate

Climate is the annual average of a location's long-term average temperature and precipitation. A region's climate is made up of the regular weather factors that prevail there for a prolonged period of time, often 30 years or more, including temperature, precipitation, humidity, and other atmospheric components.

It gives us insight into the typical weather patterns and trends in a region, enabling us to distinguish between various climatic zones like tropical, temperate, dry, etc. Climate is the proper identification of this phrase in Option (a).

The other choices are incorrect. Climate is the long-term average of a region's weather conditions. The state of the atmosphere at a specific location and time is called the weather. The technique of anticipating future weather conditions is called weather forecasting.

5. Correct option: (D) All of these

Temperature, humidity, and rainfall are a few examples of weather components. The term "weather" describes the transient atmospheric conditions and changes that take place from day to day or even hour to hour. These factors are very important in influencing the

local weather at any given time. Other weather factors may include cloud cover, visibility, wind speed, air pressure, cloud cover, and atmospheric conditions like fog or frost. Option (d) is correct as it includes all the mentioned elements of weather.

6. Correct option: (D) All of these

Forecasting the weather has many uses and benefits in daily life. By giving people early warnings and enabling them to take the required steps to preserve lives and property, it assists in preparing for adverse weather conditions, such as storms, hurricanes, or excessive rains.

It helps with daily activity planning since it allows people to plan outside events, travel, and other activities in accordance with the weather prediction. Weather forecasting is also essential for organizing agricultural activities since it allows farmers to arrange irrigation, planting, and harvesting operations depending on the expected weather. The right response is option (d), which contains all of the advantages of weather forecasting that have been highlighted.

7. Correct option: (D) All the above

The humidity of a place refers to the amount of water vapor present in the air. It is an indicator of the amount of moisture in the air and has a big impact on how comfortable or uncomfortable people feel in different weather situations. Low humidity can result in dryness, which can have an adverse effect on the health of the skin, the respiratory system, and some materials. High humidity can make the air seem heavy and stuffy.

Precipitation, or falling water from the sky, includes both rainfall and snowfall. Humidity and precipitation are not the same. Precipitation is the amount of water that falls from the sky, whereas humidity is the amount of water vapor in the air.

8. Correct option: (C) Polar region

Penguins have evolved to exist in the polar regions, especially in Antarctica and its surroundings. They are equipped to endure in severely cold and icy conditions. Penguins have various adaptations that allow them to survive in these extreme temperatures, including a coat of insulating blubber, thick feathers that trap air for further insulation, and unique glands to remove extra salt from their bodies. Their distinctive body composition and streamlined form also help them swim and dive more effectively in frigid waters. The polar area is where penguins are evolved to reside, hence option (c) is the right response.

Equatorial areas are hot and humid, whereas rainforests are warm and moist. Deserts are hot and dry. Because they are not suited to the heat or the humidity, penguins would not be able to live in these environments.

9. Correct option: (B) Rajasthan

The state of Rajasthan, which lies in the northwest of India, is renowned for its hot and dry weather. It frequently encounters summertime temperatures far beyond 40^0 Celsius (104 degrees Fahrenheit), which are considered to be severe. The area is known for its extremely low rainfall and dry and semi-arid climate. Rajasthan is among the hottest and driest places in India due to its climate. The Thar Desert, the largest desert in India, is located in Rajasthan.

The other choices are incorrect. The climates of Kashmir are chilly and mountainous, Kerala is hot and humid, and the North East has a humid, subtropical environment.

10. Correct option: (B) They cannot swim

Penguins are well-suited to live in the water and have excellent swimming abilities. Being excellent swimmers, they move through the water with tremendous agility using their wings, which have changed into flippers. Penguins spend a large amount of their lives in

the water where they forage for fish and other marine life. They are widely recognized for their swimming prowess.
The other statements are all correct. Penguins have webbed feet, streamlined bodies, and do snuggle together. Penguins keep warm by huddling together, and their webbed feet and sleek bodies enable them to swim and dive.

11. Correct option: (C) Desert regions

The adaptive traits listed in the query, such as being nocturnal, having big, cushioned feet, sweating less, and having water-saving systems, are traits frequently observed in animals living in desert environments. Animals living in deserts may survive in their environment despite the intense heat and lack of water because to these characteristics.
Numerous desert animals are nocturnal, meaning they are most active at night when it is cooler and less water evaporates. Desert animals are able to avoid sinking into the sand because to their wide, cushioned feet, which help them disperse their weight over a broader area. Because sweating causes water loss and desert animals must preserve water, they do not sweat as much. Some desert animals have unique water-saving strategies to aid them. For instance, camels can create water when necessary thanks to the fat that is stored in their hump.

12. Correct option: (A) Climate

The average weather pattern taken over a long period, such as 25 years or more, is known as "climate." The term "climate" refers to the normal atmospheric conditions, such as temperature, precipitation, humidity, wind patterns, and other meteorological factors, that are long-term fixtures in a certain area. It helps us comprehend the typical weather patterns encountered in many parts of the world and gives us a larger perspective on the long-term weather trends in a certain place.
Weather refers to the atmosphere's present condition at a specific location and time. The process through which a species or creature adapts to its environment over time is change. Humidity refers to how much water vapor is present in the atmosphere.

13. Correct option: (A) Polar region

The coldest region on Earth is the polar region, specifically the areas near the North Pole and the South Pole. Due to their proximity to the Earth's poles and the prolonged darkness of the winter, these areas suffer very frigid temperatures all year long. The main polar regions recognized for their arctic climates, ice, and snow are Antarctica and the Arctic, both close to the poles.
On the other hand, tropical locations are close to the equator and receive the most direct sunshine, which causes the temperature to rise as well. The temperate zones, which lie in between the tropical and polar regions, are more temperate than any of them. Near the oceans are the coastal zones. Coastal places often have milder temperatures than inland areas because the seas serve to regulate local temperatures there.

14. Correct option: (A) Fatty tissue

The hump of a camel is a reservoir of fatty tissue, not water or milk. The capacity of camels to store fat in their humps, which they may use as an energy reserve during periods of food scarcity, is widely documented. Contrary to common opinion, camels are extremely adept at conserving water and may go for extended periods without drinking. Instead of acting as a water storage organ, the hump. During its travels through dry and desert areas, the camel receives vital nutrition and energy from the fat tissue in the hump. The camel's bodily fluids, not its hump, are where it stores water. The kidneys of the camel are excellent at retaining water. The camel's mammary glands, not its hump, are where milk is produced.

15. Correct option: (A) Pressure

The appropriate response is (a) Pressure. Usually, a daily weather report does not include pressure. In order to convey information about the present or predicted weather conditions, the daily weather report often concentrates on factors like Temperature (b), Rainfall (c), and Humidity (d). Understanding the current atmospheric conditions and making appropriate plans for your everyday activities depends on these factors.

Although crucial for analyzing weather patterns and formulating forecasts, pressure is not often covered in a brief and uncomplicated daily weather report intended for the general audience. Instead, to evaluate and forecast more complicated weather systems, meteorologists frequently study and analyses pressure values and trends.

16. Correct option: (C) Meteorological department

The department that prepares the weather report is the meteorological department. It is in charge of tracking, evaluating, and predicting weather patterns. To provide precise and fast weather reports and predictions, the meteorological department gathers data from numerous weather stations, weather satellites, and other sources. These reports are essential for many industries that rely on weather data, including public safety, aviation, agriculture, and many others.

A nation's telecommunications infrastructure, including phone lines, cell towers, and internet cables, is the responsibility of the telecom department. A nation's electrical system, which distributes electricity to residences and businesses, is under the control of the electrical department. Buildings, bridges, and other public works projects are within the purview of the civil department.

17. Correct option: (D) Moisture

The amount of humidity in the atmosphere is a measure of the moisture content in the air. In order to evaluate weather conditions and determine how comfortable it is for people to be outside, it is crucial to know the air's humidity level. Our sense of temperature can be affected by the air feeling damp and high humidity levels. On the other side, low humidity can result in dry air, which can have an adverse effect on the skin, respiratory health, and some materials.

To determine how hot or cold something is, use the term "heat." A substance's temperature is a measure for the amount of heat it contains. Materials comprised of atoms or molecules are known as chemicals.

18. Correct option: (B) Rain gauge

The instrument used to measure rainfall is called a rain gauge. It is a simple meteorological instrument made to gather and measure the total amount of precipitation—mostly rainfall—over a predetermined time frame. Although there are many other types of rain gauges, the most popular one is a cylindrical container with markings showing the quantity of rainfall in inches or millimeters. Rain comes into the gauge, gathers in the container, and the height of the water gathered reflects the quantity of precipitation that fell during the specified time period.

An instrument used to gauge atmospheric pressure is a barometer. An instrument used to measure temperature is called a thermometer. A maximum-minimum thermometer is a device that measures the temperatures at their highest and lowest points throughout a specific time period.

19. Correct option: (C) Maximum – minimum thermometer

The maximum and minimum temperatures of a day are recorded using a maximum-minimum thermometer. A 24-hour day is typically the time span for which this sort of thermometer is intended to monitor and record the peak and lowest temperatures. Two distinct scales—one for monitoring the highest temperature and the other for the

minimum temperature—and a U-shaped glass tube make up the device. The liquid expands as the temperature rises and travels up the tube, revealing the highest temperature achieved. The liquid, on the other hand, contracts as the temperature decreases and flows down the tube, indicating the minimum temperature.
Both a Celsius and a Fahrenheit thermometer are used to monitor temperature, however neither one records the peak and lowest temperatures for a certain time period. Although a hydrogen thermometer may measure extremely low temperatures, it is uncommon to use one to note the high and low temperatures for a given day.

20. Correct option: (B) Sun

All the changes taking place in the atmosphere are primarily caused by the (b) Sun. The fundamental energy source for the Earth's weather and climate systems is the Sun. The uneven heating of the Earth's surface caused by solar radiation leads to the creation of air masses, wind patterns, and weather events. The water cycle, which produces clouds and precipitation through the evaporation and condensation of water vapor, is likewise fueled by the energy of the Sun.
The atmosphere's changes are not significantly influenced by the Moon, Earth, or stars. The tides are impacted by the Moon's gravity, although the atmosphere is not significantly affected. There are minor changes in the atmosphere brought on by the Earth's rotation, although these changes are minimal. The atmosphere is not much impacted by the stars at all.

21. Correct option: (A) Polar regions

The places where the sun does not rise for six months and does not set for the other six months are the (a) polar regions. These areas are close to the poles of the Earth, notably the North and South Poles. The tilt of the Earth's axis causes continuous darkness for roughly six months during the polar winter and continuous sunshine for about six months during the polar summer. The Earth's axis is tilted in relation to its orbit around the Sun, which affects the length of the day at various latitudes throughout the year.
This phenomenon is not present in the equatorial, northern, or southern areas. Even though the amount of sunshine varies throughout the year, the sun rises and sets in these areas every day.

22. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.
Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

23. Correct option: (D) Weather

he short-term variation that occurs in the atmosphere is called "weather." The daily variations in temperature, humidity, precipitation, wind speed, and cloud cover that can occur over short periods of time are referred to as weather. It is the state of the atmosphere at a specific location and time, offering details about the current circumstances.
Contrarily, climate refers to the typical weather patterns for a certain place over an extended period of time. Usually, the length of a climate measurement is 30 years. The quantity of water vapor in the air is known as humidity. What determines how hot or cold anything is its temperature.

24. Correct option: (B) altitude or height above sea-level

The climate at a place depends on various factors, but one of the most significant factors is altitude or height above sea-level. The atmospheric pressure drops as altitude rises, resulting in thinner air. Weather patterns and temperature are directly impacted by this alteration in atmospheric circumstances. In general, temperatures are cooler at greater altitudes, which explains why mountains and high plateaus typically have colder climates than low-lying locations.

While elements such as air pressure, wind speed, and temperature affect climate in crucial ways, the height or elevation above sea level frequently has a decisive impact on the general climatic features of a given region.

25. Correct option: (D) Polar bear

The animal that possesses blubber under its skin is the (d) Polar bear. Blubber is a thick layer of fat found under the skin of certain marine mammals, such as whales, seals, and polar bears. It serves as an insulating layer, helping these animals to stay warm in cold environments, particularly in the frigid waters of the polar regions. Polar bears, in particular, rely on their blubber to maintain body heat and survive in the Arctic's harsh and icy conditions.

There is no blubber under the skin of red-eyed frogs, elephants, or birds. In tropical jungles, there exist tiny amphibians called red-eyed frogs. In savannas and woodlands, elephants are huge creatures. Feathered creatures with wings are birds. Due to their climatic adaptations, none of these creatures require blubber to keep warm.

LEVEL-II

1. Correct option: (C) Toucan

A big, long beak is one of the distinguishing characteristics of toucans, which is known about. In addition to its amazing size, its beak has rich coloring that is frequently decorated with vivid colors. In addition to other functions, a toucan's beak enables it to reach fruits and seeds that are higher up on branches than it is physically able to reach. It also participates in communication, territorial displays, and assists in mate attraction.

Pigeons, penguins, and kiwis don't have very lengthy beaks. In order to dig for food, kiwis have long, narrow beaks. The small, stubby beaks of penguins are employed for capturing fish. Pigeons utilize their short, pointed beaks to devour seeds and other small foods.

2. Correct option: (D) Catch its prey

A sharp beak is a specialized weapon used by eagles to effectively grab and eat their prey. Eagles are raptors with exceptional hunting skills, and their beaks are built to shred and tear apart their prey's flesh. They can grip and hold onto their catch tightly thanks to their sharp, hooked beak.

Additionally, it helps the eagle swiftly break off bite-sized chunks of meat, making it simpler for it to devour. While eagles occasionally migrate, they are not known for hiding from or fleeing from predators, and their beaks are mostly specialized for their carnivorous diet and hunting ability.

They are apex predators, meaning they are at the top of the food chain and lack natural predators, therefore option A is to flee from them. Because they are huge birds, they are not particularly adept at hiding from predators, thus choose option B. Eagles migrate, option C, although their beak does not aid them in this endeavor.

3. Correct option: (D) All of these

Numerous elements, such as average temperature, soil type, and yearly precipitation, have an impact on the variety of species in varied ecosystems. Each of these elements has a significant impact on the habitat's environmental characteristics, which in turn impacts the kinds of species that may survive there. Different species have distinct adaptations

that enable them to endure and procreate in certain environmental circumstances. For instance, certain species could do better in hot, dry habitats with particular soil types, whereas others might do better in cold, damp conditions with various soil traits. The sum of these elements leads to the total biodiversity seen in different environments across the world.

4. Correct option: (B) Orchids

The term "epiphyte" refers to plants that grow on top of other plants (often trees) without taking nutrients from those plants. The most popular epiphytes are orchids. With the aid of specialized adaptations, they may adhere to tree trunks or branches and draw water and nutrients from the surrounding trash, the rain, and the air.

The popularity of orchids in horticulture and decorative plant collections can be attributed to their well-known for having an array of exquisite blossoms. The well-known tree species fir and pine do not develop as epiphytes on other plants, despite the fact that they are not epiphytes.

Both fir and pine are tree species, not epiphytes. They are rooted to the ground and develop roots as they grow in the soil.

5. Correct option: (B) Hibernation

Hibernation is the term for when animals sleep or go into a state of reduced activity in the winter to preserve energy and endure severe weather. Many animals, especially tiny mammals like bears, groundhogs, and bats, hibernate as a seasonal adaptation to withstand lower temperatures and scarcer food sources. The animal may use its stored energy stores to survive until the warmer months return since during hibernation its metabolic rate and body temperature drop dramatically to fit the environment. Torpor enables animals to endure the harsh winter months when food is in short supply.

Different methods that animals adapt to their surroundings include camouflage, movement, and mimicry. With the use of camouflage, an animal can conceal itself from predators by blending in with its environment. An animal can move about its surroundings through locomotion. Animals can mimic other organisms to look like them, frequently for trickery.

6. Correct option: (D) All of these

In addition to India, Malaysia is located in Southeast Asia, Brazil is in South America, Kenya is in East Africa, the Republic of Congo is in Central Africa, Indonesia is in Southeast Asia, Nigeria is in West Africa, and Uganda is in East Africa. Warm temperatures, high humidity levels, and year-round rainfall in these areas often provide the perfect environment for the luxuriant and diversified ecosystems seen in tropical rainforests. Numerous plant and animal species found nowhere else on Earth may be found in these jungles, which are renowned for their extraordinary biodiversity

7. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.

Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

8. Correct option: (B) Camouflage

A turtle may camouflage itself by blending in with its surroundings using the colors and patterns on its shell. The ability to blend in with their environment allows animals of

many different kinds to use camouflage, which makes it more difficult for predators to identify them and for prey to detect their presence.
In the case of turtles, the patterns and hues of their shells frequently match the forest floor or other aspects of their natural habitat, giving them some kind of defense against possible dangers. By lowering their probability of being seen by predators or prey, this adaptation boosts their odds of surviving.
Some creatures go into hibernation during the winter months, which is a condition of inactivity and low metabolic rate. Animals travel from one location to another during migration, frequently in pursuit of food or a better environment. Behavior is how an animal acts or responds to its surroundings.

9. Correct option: (C) Average in weather pattern in a few years

Climate is the long-term average of the weather in a certain area over a lengthy period of time, usually spanning decades or even centuries. It considers variables including temperature, humidity, precipitation, wind patterns, and other long-term atmospheric conditions. For a variety of uses, including agriculture, urban planning, and environmental research, climate provides important information that helps us understand the normal or predicted weather patterns in a particular place.
On the other hand, weather is the momentary condition of the atmosphere. From day to day, or even from hour to hour, it might alter. Many of the same elements that affect climate also affect weather, although it is also impacted by more regional causes like thunderstorms and hurricanes.

10. Correct option: (A) A white fur, fat below skin, keen sense of smell

Polar bears can survive in extremely cold settings because to a number of adaptations, including:
Their fur is really translucent, giving it the appearance of being white, which helps them conceal themselves when hunting and blend in with their snowy surroundings.
Polar bears have a thick layer of blubber under their skin that serves as insulation and aids in the preservation of body heat in the chilly Arctic conditions.
Polar bears have a keen sense of smell that enables them to locate seals and other prey at a distance. This skill, which enables them to find food even when it is concealed beneath the ice, is essential to their survival in their frigid habitat.
The alternative choices are incorrect. Large eyes, thin skin, and a long tail would all be problems for bears in the cold. Large eyes would make it difficult for bears to see in the snow. The bears' huge, white paws would aid in swimming but not in keeping them warm.

11. Correct option: (B) Swimming in water

Frogs and ducks have webbed feet as a form of adaptation to help them swim effectively in water. Their feet have larger surface area thanks to the webbing between their toes, enabling them to push against the water with more power. This helps them swim and move through watery surroundings. For ducks and aquatic frogs who spend a lot of time in the water, this adaption is extremely advantageous.
The other options are not correct. Frogs and ducks do not need their webbed feet to hold branches tightly, fly in the air, or grab and eat insects.

12. Correct option: (C) Adaptation

An organism's ability to grow and live in a given climate or habitat is referred to as adaptation. These characteristics are the product of natural selection and evolutionary processes, in which qualities that offer an organism a competitive edge in a particular environment is favored and handed down through generations. In order to properly obtain food, shelter, and reproduce within their ecological niche, organisms undergo adaptations to deal with elements including temperature, precipitation, humidity, and other

environmental circumstances. Therefore, an organism's fitness and success in a given environment are greatly influenced by its adaptations.
The alternative choices are inaccurate. Climate variables including temperature, humidity, and wind are referred to as the short-term weather. Climate is the collection of long-term, typical weather patterns. None of these words discuss the specific traits that an organism possesses that make it suitable to a certain environment.

13. Correct option: (A) 15 º – 40 º C

The temperature in the tropics ranges from 59 º to 104 º Fahrenheit, or 15 º to 40 ºC. Near the equator, tropical zones receive warm to hot temperatures all year long. Due to these areas' close closeness to the equator, where they receive constant, bright sunshine, the temperature there stays relatively high. Depending on the location, altitude, and other local conditions, the precise temperature within this range might change, but tropical locations often have warm climates with high humidity levels that sustain a wide variety of ecosystems and lush flora.

14. Correct option: (B) Arboreal fi burrow

Burrows and arboreal habitats are two separate categories of animal habitats. Burrowing animals dwell underground in tunnels or shelters, whereas arboreal creatures live in trees. Some creatures, like squirrels, may dwell in both burrows and trees, therefore these two habitat types are not mutually exclusive. The pair Arboreal fi Burrow is incorrect since the two phrases are not equivalent.
The other choices are the right ones. Animals classified as terrestrial or aquatic or amphibian live on land, in water, or both.

15. Correct option: (C) Long grasping tail

All the other choices (A, B, and D) are traits that are typically associated with creatures that are evolved to surviving in cold conditions, such as the Arctic or polar areas, which is why C) is the odd one out.
A) Thick layer of fat under the skin: Polar bears and seals use this adaption, known as blubber, to help them insulate themselves.
B) White fur: As a kind of camouflage, several animals have evolved white fur to assist them blend in with their environment.
D) Animals with broad, big feet and keen claws, such as polar bears, frequently exhibit this adaptation, which makes it easier for them to move around and grasp frozen surfaces and to successfully acquire prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

16. Correct option: (B) They cannot swim

Penguins have exceptional swimming skills, making them well-suited to life in the water. They are adept swimmers and use their wings, which have become flippers, to navigate through the water with great dexterity. Penguins spend a significant portion of their lives foraging for fish and other aquatic creatures in the ocean. They are also known for having excellent swimming skills.
All of the other claims are true. Penguins do cuddle up together and have webbed feet and streamlined bodies. Penguins can swim and dive thanks to their webbed feet and streamlined bodies, and they keep warm by huddling together.

17. Correct option: (C) To blend in and hide

The ability to change color allows chameleons to blend in with their environment. This enables them to ambush prey and hide from predators. They can adjust their skin color to

match the colors of nearby objects, such as leaves or branches, which helps them remain unnoticed. Chromatophores are peculiar skin cells that are present in chameleons. These cells carry pigment in a variety of colors. These cells are really growing or shrinking as a chameleon changes color. This alters how much pigment is exposed, which in turn alters the skin color of the chameleon.
The other options are not correct. Chameleons do not change color to use less energy, to change their temperature, or to help them hibernate.

18. Correct option: (A) Highest day temperature and lowest night temperature of the day

The highest recorded temperature during the day, often in the afternoon, is referenced by the maximum temperature that is shown in the weather broadcast. The highest temperature that day was represented by this.
The minimum temperature listed in the weather broadcast is the lowest reading made overnight or early in the morning, usually just before daybreak. The lowest temperature that day was represented by this.
The alternate choices are incorrect. Usually, the weather bulletin does not include the month's highest daytime or nighttime temperatures. Additionally, it is not customary to include the temperature recorded at noon and at midnight because these are not necessarily the highest and lowest temperatures of the day. The average highest and lowest temperatures of the day and night are also not often shown since they are less useful than the actual highest and lowest temperatures.

19. Correct option: (C) Red-eyed frog and beard ape

Red-eyed frogs and beard apes are both found in tropical rainforests, which are hot and humid environments. The rainforests provide these animals with the moisture and shade they need to survive.
All around the world, tropical rainforests are home to red-eyed frogs. They can see in the dark thanks to their vivid red iris. Additionally, sticky pads on the foot of red-eyed frogs enable them to climb trees.
The Western Ghats of India are the home of the bearded ape. They go by the name "lion-tailed macaques." A long, thick mane of hair in bearded apes serves as sun protection. Additionally, they have a long tail that aids in balance while climbing trees.
Elephants in Africa and polar bears can only be found in cold regions; hence they cannot live in hot and humid settings. The tundra and desert, which are both cold and dry habitats, are where you may find camels and reindeer.

20. Correct option: (C) Average weather pattern of many years

Climate is the long-term, typically spanning decades or even centuries, average of the weather in a particular place. Temperature, humidity, precipitation, wind patterns, and other long-term climatic conditions are all factors that are taken into account. Climate offers crucial information that helps us comprehend the typical or anticipated weather patterns in a certain location for a range of tasks, including agriculture, urban planning, and environmental research.
Contrarily, weather refers to the current state of the atmosphere. It might change from day to day, or even from hour to hour. The same factors that influence climate also have an impact on weather, albeit it is also influenced by more localized factors like thunderstorms and hurricanes.

21. Correct option: (D) All of these

Tropical rainforests are characterized by hot and humid climates, enormous numbers and varieties of animals, and competition for food and shelter.

The environment in tropical rainforests is often hot and humid, with average temperatures of 15 ° – 40 ° C. Additionally, there is a lot of humidity, with typical percentages ranging from 70 to 90% Huge variety and quantity of creatures: The tropical rainforests are home to a huge diversity and number of animals. Tropical rainforests are home to about half of all species on Earth, or an estimated 10 million different kinds of plants and animals.
Competition for food and shelter is a result of the high levels of biodiversity seen in tropical rainforests. These creatures may evolve specialized adaptations as a result of this intensive competition.

22. Correct option: (A) They cannot swim

Penguins can swim quite effectively, which makes them perfect for a life in the sea. They are skilled swimmers and can navigate through the water with incredible agility using their wings, which have evolved into flippers. Penguins forage on fish and other aquatic animals in the ocean for a sizable portion of their life. They are also renowned for their prodigious swimming abilities.
The other assertions are all accurate. Penguins have webbed feet, streamlined bodies, and they do snuggle up together. Penguins' webbed feet and streamlined bodies enable them to swim and dive, and they keep warm by huddling together.

23. Correct option: (C) Minimum and maximum thermometer

A unique kind of thermometer known as a minimum-maximum thermometer is used to track the daily minimum and maximum temperatures. There are two bulbs in it, one loaded with mercury and the other with alcohol. The minimum temperature is recorded using an alcohol bulb, while the highest temperature is recorded using a mercury bulb.
The temperature of a person or a material is measured using clinical thermometers or laboratory thermometers, respectively. They are not intended to keep track of the daily minimum and maximum temperatures.

24. Correct option: (D) Tropical rainforest

A tiger is a carnivorous animal that fits the definition; it is distinguished by its striking stripes and extraordinary speed and agility when pursuing prey. In Asia, grasslands and tropical rainforests are the main habitats of tigers. They are strong predators that have evolved to live in these luxuriant, thick forest habitats. These animals have stripes on their body that help them to camouflage in the dense vegetation of the rainforest. They also have sharp claws and teeth that help them to catch their prey.
The Arctic, which is significantly dissimilar to the tropical jungle in terms of habitat, is where polar bears may be found. Tropical rainforests are often very hot and humid, while deserts are typically very hot and dry. Ocean-dwelling animals have gills or a thick covering of blubber as adaptations to help them survive in the saline water.

25. Correct option: (C) Long grasping tail

The qualities of all the other options (A, B, and D) are often associated with animals that have evolved to survive in frigid climates, such the Arctic or polar regions, which makes C) the odd one out.
Polar bears and seals employ this adaptation, known as blubber, to help them insulate themselves. A) Thick layer of fat under the skin.
B) White fur: Several animals have developed white fur as a form of camouflage to help them blend in with their surroundings.
D) This adaptation is typically seen in animals with large, broad feet and sharp claws, such as polar bears, which makes it simpler for them to walk about, grab icy surfaces, and effectively catch prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for

support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

26. Correct option: (C) Mix with the surroundings

Tigers and leopards have evolved to have stripes on their body, which gives them the ability to blend in with their environment and serve as an excellent kind of camouflage. Their ability to blend in and approach their target covertly is especially helpful for their hunting tactics.

In the dappled light and shadows of their native environments, such as grasslands and woodlands, the stripes blur the shape of their bodies, making it more difficult for their prey to identify them. Their ability to blend in with their surroundings and approach their victim undetected until they are prepared to strike is crucial to their effectiveness as predators.

27. Correct option: (A)

28. Correct option: (D) a – iv; b – ii; c – i; d – iii

a. Red-eyed frog has iv. Sticky pads on feet. They can climb trees because to this adaptability. Arboreal creatures, red-eyed frogs spend the most of their time in trees. Their feet include sticky pads that allow them to effortlessly move around in trees and hold onto branches.

b. Penguin has ii. Streamline body. This modification enhances their capacity for swift and effective swimming. Penguins can hunt fish and other prey using their streamlined bodies since they are skilled divers and swimmers.

c. Tiger has i. Very sensitive hearing. They have the capacity to hear noises that are imperceptible to humans. This modification makes it easier for them to discover prey and recognize predators.

d. Lion-tailed macaque has iii. Silver-white mane. They can attract mates thanks to this modification. Additionally, the main makes lions appear bigger and more frightening, which can aid in fending off predators.

So, the answer is D. a – iv; b – ii; c – i; d – iii.

29. Correct option: (A) a – iv; b – iii; c – ii; d – i

a) Elements of weather - iv) Temperature, humidity: The term "elements of weather" refers to a variety of elements, such as temperature and humidity, that make up the atmospheric conditions.

b) Cause of all changes - iii) Sun: The sun is the main influence on Earth's weather and climate. All alterations in weather patterns and atmospheric conditions are influenced by its energy.

c) Tropical region - ii) Hot and wet: High temperatures and high humidity combine to create a hot and humid environment in tropical locations.

d) Desert climate - i) Hot and dry: Low humidity and high temperatures create a dry and warm atmosphere in desert settings.

30. Correct option: (B) (i), (iii)

The conditions from the above list that are responsible for the presence of a large number of plants and animals in tropical rainforests are:

(i) Humid and hot climate: Tropical rainforest’s humid and hot climates are ideal for the development and spread of a wide range of plant species. High temperatures and humidity encourage quick plant development, which results in a varied and thick forest canopy.

(iii) Abundant rainfall: Tropical rainforests get an abundance of rainfall, which guarantees a steady supply of water, which is essential for the survival and development of both plants and animals. It supports a wide range of plant growth and offers home to a large number of animal species.

3. Our Forest

Solutions

LEVEL-I

1. Correct option: (B) Sunderlal Bahuguna

The Chipko movement was started by Sunderlal Bahuguna in the Garhwal forests. The movement was aimed at stopping the felling of trees for commercial purposes. The movement was non-violent and the villagers hugged trees to stop them from being felled.

2. Correct option: (D) 20 %

The percentage of forest area of our county is 21.71%. The total forest cover in India is 80.9 million hectares (x 100 square kilometres) which makes 21.71 percent of the total geographical area in 2021. According to a recent study, India's forest/tree cover has increased by more than 1,500 sq.

3. Correct option: (C) Both A and B

The Chencho's hunt wild animals like boar and deer, but with the increasing interest in wildlife conservation, they are content to hunt small animals like lizards, rabbits and wild birds.

4. Correct option: (A) Anantha Giri Forest

Konda Mallipudivana Samrakshana Samithi is located at the foothills of Anantgagiri forest.

5. Correct option: (D) All of the above

Dandakaranya forest holds importance due to its mention in Ramayana. The location housed various demons and deadly creatures. It is also believed that Hindu deities lived there. Panchavati forest is present in Nashik city, on the banks of the Godavari River. Nallama forest is located in Andhra Pradesh and Telangana. The forest is part of Eastern Ghats.

6. Correct option: (B) Conservation

Nature conservation is the moral philosophy and conservation movement focused on protecting species from extinction, maintaining and restoring habitats, enhancing ecosystem services, and protecting biological diversity.

7. Correct option: (A) Fauna

Total number of animal species present in a region are called fauna. Wildlife refers to animals that cannot be domesticated by human beings and known as 'fauna'. India is also home to a variety of wildlife. Tigers are found throughout India - from the Himalayas to Kanyakumari. The Royal Bengal Tiger is our national animal.

8. Correct option: (C) Soil erosion

Soil erosion is the denudation or wearing away of the upper layer of soil. It is a form of soil degradation. This natural process is caused by the dynamic activity of erosive agents, that is, water, ice, snow, air, plants, and animals.

9. Correct option: (D) Forest

The plants release oxygen through the process of photosynthesis. The plants help to provide oxygen to animals for respiration. They also maintain the balance of oxygen and carbon dioxide in the atmosphere. That is why forests are called green lungs.

10. Correct option: (B) Rosewood

Rosewood is a species that can be used in projects to restore degraded lands, which can be applied to thousands of hectares of degraded lands across the region impacted by extractive activities. Moreover, the trees planted in this project can serve as a seed bank for others who may wish to reforest and conduct ecological restoration that includes rosewood, either for strict conservation or for managed, non-destructive leaf harvest, which could generate additional income for local communities.

11. Correct option: (A) Forest of cold areas
Alpine forests are mostly found in mountainous areas. Alpine forests can be found across the Himalayas, at elevations ranging from 2,900 to 3,500 m or indeed up to 3800 m above sea level, based on location and plant variety.
12. Correct option: (D) All of the above
overpopulation, urbanisation and Industrialisation are the causes for deforestation.
13. Correct option: (D) All the above
Limiting the use of firewood, using recycled paper and using biofuels for cooking all will result in preventing deforestation
14. Correct option: (A) Biodiversity
Biodiversity is the variety of plant and animal life in a particular habitat.
15. Correct option: (D) Resources
Materials present in large quantities for future needs are called Resources. A resource is a physical material that humans need and value such as land, air, and water. Resources are characterized as renewable or non-renewable; a renewable resource can replenish itself at the rate it is used, while a non-renewable resource has a limited supply.
16. Correct option: (A) 25Km
17. Correct option: (D) all the above
Elephants, rabbits and monkey etc are found in the forestry of Adilabad district.
18. Correct option: (D) all the above
Writing pads, cot and pencil are made up from wood.
19. Correct option: (D) all the above
We all depend on the forest for groundwater, rainfall and medicinal plants.
20. Correct option: (D) kerosene
Kerosene can be produced from distillation of crude oil (straight-run kerosene) or from the cracking of heavier petroleum streams (cracked kerosene).

LEVEL-II

1. Correct option: (B) Crown
The branchy part of a tree above the stem is known as the crown
2. Correct option: (C) Both A and B
A Forest is a purifier of water and air because it helps in bringing rain and helps in maintaining ground water and provides fresh water. Moreover, forests are responsible for maintaining a balance between oxygen and carbon dioxide gas in the atmosphere and provide fresh air.
3. Correct option: (C) Food chain
The interdependence of animals for food is called a food chain. For example, one kind of interdependent relationship would be the bees and butterflies that help pollinate many kinds of flowers or the bats and birds that carry pollen to other areas so that new plants can grow. Even a predator-prey relationship is a form of interdependence.
4. Correct option: (D) honey
Honey is produced from bees (Yes, insects are considered animals) and therefore is not a vegan product. According to PETA, honey is the energy source for bees and they would starve without it, especially during winter and poor weather conditions. Conventional beekeeping works to harvest the largest amount of honey.
5. Correct option: (B) humus
Decomposers feed on dead plants and animal tissues. They convert them into humus. This activity also helps in increasing the soil fertility.

6. Correct option: (C) Porcupine
Porcupines are large rodents with coats of sharp spines, or quills, that protect them against predation. Sal, teak and kachnar are trees.
7. Correct option: (C) Decomposers
Nature has its own recycling system: a group of organisms called decomposers. Decomposers feed on dead things: dead plant materials such as leaf litter and wood, animal carcasses, and faces.
8. Correct option: (D) All of these
Understoreys are formed due to different types of crowns, different size of crown, different height of trees. The lower layer of the forest formed due to different types of crowns in the forest is known as the understory. The constituents of the understorey are herbs and shrubs. The plants which are usually short and have green, tender stems are known as herbs.
9. Correct option: (B) Canopy
Tall trees which look like roof over other plants in the forest is called canopy
10. Correct option: (C) Porcupine
Porcupines are large rodents with coats of sharp spines, or quills, that protect them against predation. Sal, teak and kachnar are trees.
11. Correct option: (A) Bird, rabbit, elephant
Primary consumers are a group of organisms that are placed in the second trophic level. They feed on producers i.e grass and green plants. They are therefore also known as herbivores.
12. Correct option: (A) Cinchona
Cinchona bark contains quinine, which is a medicine used to treat malaria.
13. Correct option: (C) Furniture industry
Industries which are totally dependent upon the products of forests are - Paper Industry, Match Industries, Plywood Industries and Fibre Board Industries.
14. Correct option: (B) Decomposing
Several beetles and grubs were feeding on the heap and a bunch of seedlings was sprouting. “These seedlings are of the herbs and shrubs the animals also disperse the seeds of certain plants and help the forest to grow and regenerate. The decaying animal dung also provides nutrients to the seedlings to grow”, said Prof Ahmad.
15. Correct option: (C) Recycling of paper bags
Recycling of cardboard cartons and paper bags will lead to a decrease in the level of tree cutting and hence conserve forests.
16. Correct option: (B) Forest
Forest is a natural absorber of rain water. It allows rain water to seep through. The roots of the trees absorb the water and prevent it from flowing away. The trees present in the forest prevents the rain from directly hitting the ground which may result in floods. It rather hits the ground slowly. Hence, trees in the forest slow down the movement of running water and thus help in preventing floods.
17. Correct option: (C) Humus
Decomposers or saprotrophs recycle dead plants and animals into chemical nutrients like carbon and nitrogen that are released back into the soil, air, and water. Fungi like mushrooms, mildew, Mold, and toadstools can't make their own food.
18. Correct option: (B) Thermocol
Thermocol is not prepared from the wood obtained from the forest. Paper, matchsticks, and plywood are prepared from the wood.
19. Correct option: (B) Global Warming
If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.

20. Correct option: (A) Grass → insects → frog → snake → eagle
Grass is a primary producer which is consumed by insects, insects are further eaten by amphibians like frogs which are consumed by snakes, and snakes are consumed by eagles.
21. Correct option: (C) Creating flood conditions
Forest is a natural absorber of rain water. It allows rain water to seep through. The roots of the trees absorb the water and prevent it from flowing away. The trees present in the forest prevents the rain from directly hitting the ground which may result in floods. It rather hits the ground slowly. Hence, trees in the forest slow down the movement of running water and thus help in preventing floods.
22. Correct option: (B) plants and animals in a forest are not dependent on one another.
Animals are dependent on plants for shelter and food, while plants are dependent on animals for pollination and seed dispersal. The plants filter water and make it clean, and animals require clean water for drinking purposes.
23. Correct option: (D) All the above are correct.
Absence of forest will result in no food and shelter for animals and If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.
24. Correct option: (A) Amount of CO2 in air will increase
If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.
25. Correct option: (B) Dodo
The dodo (*Raphus cucullatus*) is an extinct flightless bird that was endemic to the island of Mauritius, east of Madagascar in the Indian ocean. It became extinct in 14th century from Mauritius because of large-scale hunting. Dodo was helpful in pollinating and propagating seed of this tree species. Subfossil remains show the dodo was about 1 metre (3 ft 3 in) tall and may have weighed 10.6–21.1 kg (23–47 lb) in the wild.
26. Correct option: (A) Semal and hornbill
Semal (Silk Cotton) tree is a type of native cotton tree with large red flowers. It grows rapidly to attain height up to 30 m and a girth of 3-4 m and the largest hornbill species in India, Great Hornbills are also called Great Indian or Great Pied Hornbills.
27. Correct option: (D) None of the above.
The plants are called producers, the animals are called consumers and Saprotrophs are commonly known as decomposers are correct.
28. Correct option: (D) None of the above.
The branchy part of a tree above the stem is known as the crown of the tree, the layer of vegetation in a forest which is just below the canopy is called under story and the uppermost branches and leaves of tall trees which act like a roof over the forest ground is called canopy.
29. Correct option: (B)
a – iv) The forest canopy is a structurally complex and ecologically important subsystem of the forest.
b – iii) Forest is a renewable resource. But, each year, the earth loses about 36 million acres of forest.
c – ii) Plants gives out oxygen.
d – i) Small trees are Under storeys trees are shade tolerant species, about 15 to 50 feet in height, that grow under the crowns of larger trees.
30. Correct option: (D)
a – iii) Herbs are grasses.
b – i) Watermelon is a creeper.
c – v) Pea plants are climber.
d – ii) Hibiscus is a shrub.
e – iv) Banyan is a tree.

4. Conservation of Plants and Animals

Solutions

LEVEL-I

1. Correct option: (C) Endangered Species

The species very near to extinction, if conservational measures are not prompt are called endangered species.

2. Correct option: (A) International union for conservation of nature and natural resource

The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of: 1. Endemic plant and animal species present in the biodiversity hotspots.

3. Correct option: (C) Both A and B

Wildlife is conserved by both in situ and ex situ methods.

In situ or on-site conservation involves the conservation of genetic resources of natural populations of plant and animal species, such as from forests, National Parks, and Biosphere Reserves.

Ex situ conservation involves the preservation of components of biological diversity outside their natural habitats, such as in Zoological Parks.

4. Correct option: (B) 50000

India has more than 50000 genetically different strains of rice.

5. Correct option: (A) Biosphere reserve

Biosphere reserves are large areas of protected land for conservation of wildlife, plant and animal resources and traditional life of the tribals living in the area. The biosphere reserves help to maintain the biodiversity and culture of that particular area.

6. Correct option: (D) All

The fundamental purpose of plant gene banks is to preserve genetic varieties in the form of seeds or cuttings for plants that reproduce vegetatively, and subsequently, make them accessible for use in plant breeding and research in the future.

7. Correct option: (C) Habitat destruction

Habitat loss is the primary cause of higher extinction rates. Other causes include habitat changes, over-exploitation of wildlife for commercial purposes, the introduction of harmful nonnative species, pollution, and the spread of diseases.

8. Correct option: (D) Threatened species

The IUCN Red List provides information about the geographic range, population size, habitat and ecology, use and trade, threats and conservation actions for species.

9. Correct option: (B) Fossil Fuel

Coal, natural gas and crude oil are called as fossil fuels. They are regarded as non-renewable source because it takes millions of years to form and once these resources are used, they cannot be regained.

10. Correct option: (C) Increasing Humidity and rainfall

Forests helps in producing rainfall. More are the trees; the more water gets absorbed into clouds and the more it helps in rain falls. The roots of a tree make gaps in the soil so that when it rains the water can sink in before being absorbed by the soil. These spaces also allow air and water to reach the roots of the plants. If the forests disappear, there will be less rain resulting in dryer conditions, that eventually lead to drought. Once this happens, the damage is extensive making it difficult to plant more trees or plants later because the soil is so hard and dry.

11. Correct option: (A) Jatropha

Biofuels are of two types in which bioethanol is one of the types. Jatropha plant is used as an alternative to produce bioethanol. A high amount of energy is produced by Jatropha as its seeds are rich in fatty acid methyl ester. Hence the biofuel bioethanol can be yielded by processing the oil obtained from Jatropha seeds. Its oil is mixed with pongamia oil to produce bioethanol.

12. Correct option: (C) East Africa

Lake Victoria, also called Victoria Nyanza, largest lake in Africa and chief reservoir of the Nile, lying mainly in Tanzania and Uganda but bordering on Kenya.

13. Correct option: (C) Lungs of Planet

Amazonian rain forest in south America has the greatest biodiversity on earth; it harbors about40000 species of plants, 1,25,000 species of insects, 3000 species of fishes, 427 of amphibians, 378 of reptiles, 1300 of birds & 427 of mammals.

14. Correct option: (A) Endemic Species

Endemic species are plants and animals that exist only in one geographic region. Species can be endemic to large or small areas of the earth: some are endemic to a particular continent, some to part of a continent, and others to a single island.

15. Correct option: (D) Botanical Garden

These include aquariums, botanical gardens, gene banks etc. In Situ conservation refers to the conservation of the plants and animals in their own habitat. These include biosphere reserves, wildlife sanctuary and national parks.

16. Correct option: (A) Endemic population

The population limited to a particular geographical area is called an endemic population.

17. Correct option: (C) Root of rauwolfia serpentine

Reserpine is an indole alkaloid extracted from Rauwolfia serpentine roots, an Indian climbing shrub.

18. Correct option: (C) Gene bank

Endangered plant species are conserved through Gene bank. By this method, different genetic diversity is preserved and thus creates a genetic pool of different genetic material that acts as a resource house.

19. Correct option: (C) Cryopreservation

Germplasm conservation at liquid nitrogen temperature is called cryopreservation. It is an example of Ex-situ conservation. Ex-situ conservation is done by preserving the components of biological diversity outside their natural habitats.

20. Correct option: (A) Pollution

UNESCO's Biosphere Reserves face threats such as habitat loss, climate change impacts, invasive species, pollution, unsustainable resource use, and development pressures.

21. Correct option: (B) In situ conservation

The conservation of species in their natural habitat or natural ecosystem is known as in-situ conservation. In the process, the natural surrounding or ecosystem is protected and maintained so that all the constituent species (known or unknown) are conserved and benefited.

22. Correct option: (D) 29 December

From its creation by the Second Committee of the UN General Assembly in 1993 until 2000, it was held on December 29 to celebrate the day the Convention on Biological Diversity went into effect. On December 20, 2000, the date was shifted to commemorate the adoption of the Convention on May 22, 1992, at the Rio de Janeiro Earth Summit, and partly to avoid the many other holidays that occur in late December.

23. Correct option: (D) −196°C

Cryopreservation is the conservation of living cells, tissues organs, and microorganisms at ultralow temperature (usually that of liquid nitrogen, −196°C).

24. Correct option: (B) Maximum biodiversity

Biodiversity hotspots are the regions of megadiversity. These areas are rich in biodiversity. They reserve the endangered species.

25. Correct option: (B) Deforestation

Deforestation is the depletion of forest resources. Its main cause is the explosion of human and livestock population with the increased demand requirements of the basic needs. The major effect of deforestation is the loss of precious wildlife, rare species of flora and fauna. Directly or indirectly, deforestation caused intensified soil erosion, accentuated floods, drought and the worst pollution.

26. Correct option: (C) Contains diverse genetic resources

All life forms should be conserved because, they maintain diverse genetic resources.

27. Correct option: (B) 2.4%

India occupies 2.4% of the world's land area with its 3.28 million square kilometre area.

28. Correct option: (D) Coexistence

There are four major causes collectively called as the evil quarter- i) Overexploitation, ii) Loss of habitat, iii) Extinction of species, iv) Introduction of exotic species. Thus Co – Existence is not included in the Evil Quartet for accelerated rates of species extinction.

29. Correct option: (B) Alkaloids

Drugs obtained from plants are mostly alkaloids in nature. Examples: - Quinine obtained from the bark of Cinchona stem. Taxol obtained from the bark of Pacific yew. Reserpine obtained from Rauvolfia serpentina. Morphine obtained from the poppy plant.

30. Correct option: (D) Seed Bank

Seed banks are the conservation site where seeds are stored for conserving species as seeds can stay in a dormant stage for a longer span until it gets favorable conditions. As plants are conserved by this method outside their habitat, it is a type of Ex-situ conservation.

LEVEL II

1. Correct option: (B) Increased resistance to environmental perturbance

Biodiversity loss is defined as the extinction of species worldwide, as well as the local reduction or loss of species in a certain habitat, resulting in a loss of biological diversity. Loss of biodiversity may lead to decline in plant production, increased variability in plant productivity, Increased variability in water use and in pest and disease cycle. There would also be a decrease in resistance to environmental disturbances.

2. Correct option: (C) 1952

The National Forest Policy in India was enforced in 1952 by the government of India.

3. Correct option: (D) Red Data Book

The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of: 1. Endemic plant and animal species present in the biodiversity hotspots.

4. Correct option: (A) Soil erosion

Global warming is the rise in the temperature of the earth's atmosphere. This results in the melting of polar glaciers, which increases the sea level, causing floods in the coastal areas. Global warming can also cause an increase in the temperature of the ocean, which causes the death of marine flora and fauna. Global warming has no effect on soil erosion.

5. Correct option: (A) 1992

The United Nations Conference on Environment and Development (UNCED), also known as the 'Earth Summit', was held in Rio de Janeiro, Brazil, from 3-14 June 1992.

6. Correct option: (D) All of these

A wildlife sanctuary is an area where animal habitats and their surroundings are protected from any sort of disturbance. The capturing, killing, and poaching of animals is strictly prohibited in these regions. They aim at providing a comfortable living to the animals.

7. Correct option: (B) On 1 April 1973

The Government of India launched the Project Tiger with the aim of saving the steadily declining population of tigers in India on 1 April 1973. Project Tiger has been the largest species conservation initiative of its kind in the world.

8. Correct option: (A) Bori and Pachmarhi

The Panchmarhi Biosphere Reserve consists of one National Park named Satpura and two wildlife sanctuaries named Bori and Panchmarhi.

9. Correct option: (C) Wildlife sanctuary and National Park

Wildlife sanctuary and national parks are the places meant for conservation of biodiversity in their natural habitat.

10. Correct option: (A) Pachmarhi Biosphere Reserve

The biodiversity found in the Panchmarhi Biosphere Reserve is similar to that found in the upper Himalayan peaks and lower Western Ghats.

11. Correct option: (D) All of the above

Forests provide us with oxygen, shelter, jobs, water, nourishment and fuel.

12. Correct option: (C) Is occurring at a faster rate

Unlike previous extinction events caused by natural phenomena, the sixth mass extinction is driven by human activity, primarily (though not limited to) the unsustainable use of land, water and energy use, and climate change, and is occurring at relatively higher rates than the previous ones.

13. Correct option: (A) Habitat loss and fragmentation

When a habitat is destroyed, the carrying capacity for indigenous plants, animals, and other organisms is reduced so that populations decline, sometimes up to the level of extinction. Habitat loss is perhaps the greatest threat to organisms and biodiversity.

14. Correct option: (D) Biodiversity

Biodiversity refers to the variety of organisms existing on the earth, their interrelationships, and their relationship with the environment. Biodiversity or Biological diversity is a term that describes the variety of living beings on earth. It helps in maintaining the balance of the ecosystem.

15. Correct option: (A) Endemic species

Endemic species are plants and animals that exist only in one geographic region. Species can be endemic to large or small areas of the earth: some are endemic to a particular continent, some to part of a continent, and others to a single island.

16. Correct option: (B) Preservation of species

Creation of wildlife reserves and enforcement of laws are conservation measures that promote increased preservation of species.

17. Correct option: (C) Hunting and Poaching

The illegal hunting and poaching of animals can lead to a decrease in their numbers, which can have negative effects on the overall ecosystem. Poaching of endangered species such as elephants, tigers, and rhinos can lead to their extinction, causing irreversible damage to the environment.

18. Correct option: (D) All of the above

The species which no longer exist anywhere on the earth are called extinct species. The species which are facing the risk of extinction are called endangered species. Red data book keeps a record of all the endangered plants and animals or the species which are at continuous threat of becoming extinct. There are various colour codes for various types of species depending on the level of risk they are in when it comes to becoming extinct.

19. Correct option: (A) Habitat Destruction

Habitat loss is the primary cause of higher extinction rates. Other causes include habitat changes, over-exploitation of wildlife for commercial purposes, the introduction of harmful nonnative species, pollution, and the spread of diseases.

20. Correct option: (C) Very cold

Migratory birds fly to far away areas every year during a particular time because of climatic changes. They fly for laying eggs as the weather in their natural habitat becomes very cold and inhospitable.

21. Correct option: (D) Interbreeding

A species is a group of populations which are capable of interbreeding to produce a fertile offspring.

22. Correct option: (D) Habitat

A place where an organism or a community of organisms live, including the living and non-living conditions of their surrounding environment, is called a habitat.

23. Correct option: (A) Animals move freely inside a national park

In any national park, the animals and plants are conserved in their natural habitat. The wild animals roam freely. Hence, tourists must watch them through a well-protected vehicle for their own safety.

24. Correct option: (B) Decrease in floods

Increase in groundwater level. An increase in groundwater level is not a consequence of deforestation. It is a very broad term, which consists of cutting of trees including repeated lopping, felling, and removal of forest litter, browsing, grazing, and trampling of seedlings.

25. Correct option: (B) Biosphere

Part of the earth which supports life where living beings exists is called biosphere. The biosphere is the layer of the planet Earth, where life exists.

26. Correct option: None are correct (explanation below)

According to IUCN Red list, Labrador duck is an extinct species. Cheetah and leopard are vulnerable species whereas Jaguar is near threatened species. Hog dear, red panda, Asian elephant and blue whale are endangered species.

27. Correct option: (A) Dodo

Dodo is an extinct flightless bird that was endemic to the island of Mauritius, east of Madagascar in the Indian Ocean. It became extinct due to over-exploitation by human beings.

28. Correct option: (B) a-ii; b-iii; c-i

Dodo is an extinct flightless bird that was endemic to the island of Mauritius. Giant squirrels are Endemic to Western, Eastern Ghats and Satpura range of India. Tigers have lost an estimated 95% of their historical range. Their habitat has been destroyed, degraded, and fragmented by human activities. Hence, they are categorised as endangered animals by IUCN.

29. Correct option: (D) 1-b; 2-d; 3-a; 4-c

Giant squirrels are Endemic to Western, Eastern Ghats and Satpura range of India. Cutting of trees in the forests is known as deforestation. A food chain is a linear sequence of organisms through which energy and nutrients pass as one organism eats another. Pachmarhi comprises three protection sites: the Bori Sanctuary, Satpura National Park and Pachmarhi Sanctuary – otherwise known as the Satpura Tiger Reserve.

30. Correct option: (D) (i), (ii), (iii), (iv)

Deforestation is removal, decrease or deterioration of forest cover of an area. It causes shrinkage of fuelwood, reduced timber supply, change in climate, flash floods, soil erosion, siltation, cyclones, drought, desertification, reduced rainfall and destruction of habitat.

5. Pollution of Air and Water (Natural Resources)

Solutions

LEVEL-I

1. Correct option: (C) 6 months

Every PUC certificate is valid for a period of 6 months. Therefore, every 6 months, you need to get your car checked and re–tested for pollution emission levels to get a fresh PUC certificate issued.

2. Correct option: (A) 21%

Earth's atmosphere is composed of about 78 percent nitrogen, 21 percent oxygen, 0.9 percent argon, and 0.1 percent other gases. Trace amounts of carbon dioxide, methane, water vapor, and neon are some of the other gases that make up the remaining 0.1 percent.

3. Correct option: (C) Carbon dioxide

In the decomposition process, different products are released: carbon dioxide (CO_2), energy, water, plant nutrients and resynthesized organic carbon compounds.

4. Correct option: (C) Visakhapatnam

NTPC Simhadri Super thermal power plant at Parawada, Visakhapatnam, Andhra Pradesh.

5. Correct option: (C) Nuclear power

Nuclear energy originates from the splitting of uranium atoms – a process called fission. This generates heat to produce steam, which is used by a turbine generator to generate electricity. Because nuclear power plants do not burn fuel, they do not produce greenhouse gas emissions.

6. Correct option: (D) 1984

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

7. Correct option: (C) Chemicals

Industrial waste exhaust gases contain large amounts of toxic sulfurized gases such as SO_2, H_2S, CS_2, and sulphur-containing organic compounds. Removal of these gases from exhaust gases is one of the major applications of activated carbons.

8. Correct option: (D) All

Thermal power plant emissions cause asthma, Chronic Obstructive Pulmonary Disease Stunts lung growth, lung cancer, cardiac disease and congestive heart failure

9. Correct option: (A) PAN

Peroxyacetyl nitrate (PAN) is a secondary pollutant because it is formed in the atmosphere after the emission of primary pollutants which is a part of smog that causes air pollution.

10. Correct option: (B) Yamuna

The Taj Mahal was built on the banks of the river Yamuna, between 1631 and 1643, by the Mughal emperor Shah Jahan (r. 1628-58) as a mausoleum for his beloved wife Arjumand Banu Begum, who was also known by the title Mumtaz Mahal.

11. Correct option: (C) Mercury

The source and transmission mode are contaminated fish and shellfish. The etiologic agent is methylmercury, a by-product of acetaldehyde production, which was discharged from the Chisso factory from 1932 until 1968.

12. Correct option: (A) 31%

Forests cover 31 percent of the world's land surface, just over 4 billion hectares. (One hectare = 2.47 acres.) This is down from the pre-industrial area of 5.9 billion hectares.

13. Correct option: (D) All the above

Refrigerators, air conditioners, different types of sprays, fire extinguishers, and paints released chlorofluorocarbons in the air. And this emission contributed to the depletion of the Ozone layer.

14. Correct option: (B) July

Van Mahothsav or Vanamahothsava, lit. 'Forest festival', is an annual one-week tree-planting festival in India which is celebrated in the first week of July.

15. Correct option: (D) CNG

Petrol, diesel, and kerosene are petroleum products and they produce pollutants like carbon-dioxide and smoke. CNG or compressed natural gas is a smoke-free gas and does not spread pollution, and thus used in our vehicles. Therefore, CNG is considered as an eco-friendly fuel.

16. Correct option: (A) Nitrates and phosphates

Addition of phosphates and nitrates/fertilizers into water leads to nutrient enrichment which is also called as eutrophication. It is the process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen. It results in the loss of biodiversity.

17. Correct option: (A) CFCs

Gaseous CFCs can deplete the ozone layer when they slowly rise into the stratosphere, are broken down by strong ultraviolet radiation, release chlorine atoms, and then react with ozone molecules.

18. Correct option: (B) CO_2 and SO_2

Acid Rain is slightly acidic because it contains dissolved carbon dioxide (CO_2). Sulphur dioxide (SO_2) and Nitrogen oxides (NOx) which are normally present in the air.

19. Correct option: (A) Electrostatic precipitator (ESP)

Electrostatic precipitators are installed to control the emission of Suspended Particulate Matter (SPM). These precipitators can remove over 99 percent of the particulate matter present in the exhaust from a thermal power plant.

20. Correct option: (C) Methyl Isocyanate

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

LEVEL-II

1. Correct option: (D) All of these

Instead of the fossil fuels for our energy requirements we can use Solar Energy, Hydropower and Wind energy as there is least likelihood of release of greenhouse gases into the atmosphere.

2. Correct option: (C) 21%

Earth's atmosphere is composed of about 78 percent nitrogen, 21 percent oxygen, 0.9 percent argon, and 0.1 percent other gases. Trace amounts of carbon dioxide, methane, water vapor, and neon are some of the other gases that make up the remaining 0.1 percent.

3. Correct option: (B) Nitrogen

Sulphur dioxide, carbon dioxide, and methane are directly linked with global warming, as these gases increase heat within the aerial blanket surrounding the earth. But nitrogen is not responsible for global warming, as it is 78% in the atmosphere with no heat trapping history.

4. Correct option: (A) Infrared radiations

Molecules of carbon dioxide (CO_2) can absorb energy from infrared (IR) radiation. The energy from the photon causes the CO_2 molecule to vibrate.

5. Correct option: (A) Replace

The 3R's mantra that is used for the conservation of water includes reduce, reuse and recycle. Thus, replace is not a way to conserve water.

6. Correct option: (C) DDT

DDT and its metabolites are readily adsorbed onto sediments and soils, which can act both as sinks and as long-term sources of exposure. Because of its strong tendency to be adsorbed onto surfaces, most DDT that enters water is and remains firmly attached to soil particles.

7. Correct option: (B) Distillation

Distillation procedures will give us water free from all impurities. The distillation process removes almost all impurities from water. Distillers are commonly used for removing nitrate, bacteria, sodium, hardness (dissolved solids), most organic compounds, heavy metals and radionuclides from water.

8. Correct option: (C) Drip irrigation

Drip Irrigation system helps water to reach roots effectively. The most suitable method of irrigation for areas having low rainfall and strong winds is drip irrigation, as water will be suitably used in drip irrigation with high efficiency in low rainfall and strong wind areas.

9. Correct option: (D) Chlorination

Chlorination is a commonly used chemical method for purifying water. It kills germs in the water.

10. Correct option: (A) Air Pollution

Sulphur dioxide and nitrogen dioxide are pollutants emitted into the atmosphere as a result of the combustion of fossil fuels such as diesel and coal. Sulphur dioxide, NO_2 emissions, and primarily carbon-based particles have corroded and damaged the Taj's beautiful white exterior, leaving it with a yellow sheen.

11. Correct option: (A) Nitrogen dioxide

In the lung, nitrogen dioxide (NO_2) hydrolyzes to nitrous acid (HNO_2) and nitric acid (HNO_3), which can then cause chemical pneumonitis and pulmonary edema. Because NO_2 is poorly water soluble, it hydrolyzes more slowly than other water-soluble gases, resulting in deep lung injury in the bronchioles and alveoli.

12. Correct option: (B) CFCs

Many refrigerants, such as chlorofluorocarbons (CFCs) damage the ozone layer, while others are extremely potent greenhouse gases. In fact, one kilogram of the refrigerant R410a has the same greenhouse impact as two tonnes of carbon dioxide, which is the equivalent of running your car for six months.

13. Correct option: (C) The sun’s UV rays

The stratospheric ozone layer is Earth's “sunscreen” – protecting living things from too much ultraviolet radiation from the sun.

14. Correct option: (A, B, C) CO_2, CO, CH_4

CO_2 (Carbon dioxide) is an air pollutant, primarily due to its role in contributing to climate change when present in excessive amounts. CO (Carbon monoxide) is an air pollutant that can be harmful when present in high concentrations, as it can interfere with the body's ability to transport oxygen in the blood. CH_4 (Methane) is another air pollutant that contributes to climate change when released into the atmosphere, as it is a potent greenhouse gas.

15. Correct option: (A) Swimming in the pond

Swimming in the pond will generate the least water pollution as no harmful chemicals or pollutants would get released in the process.

16. Correct option: (B) Leakage of poisonous gas

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

17. Correct option: (C) Water Pollution

The addition of harmful materials to water that reduces its quality is called water pollution.

18. Correct option: (C) Reduced oxygen carrying capacity of RBCs

This is because when carbon monoxide and oxygen both are available to RBCs then the RBCs will bind 200 times faster with carbon monoxide than oxygen, thus forming carboxyhaemoglobin rather than oxyhaemoglobin. This carboxyhaemoglobin reduces the oxygen carrying capacity of RBCs (since RBCs contain haemoglobin, a red-coloured protein).

19. Correct option: (D) Large amount of salt

Seawater contains a large amount of salt in it. Drinking a large amount of seawater makes a person dehydrated. Hence, it is not good for health.

20. Correct option: (B) Reduce Ganga water pollution

The Ganga Action Plan or GAP was a program launched by Rajiv Gandhi in April 1986 to reduce the pollution load on the river. But the efforts to decrease the pollution level in the river became more after spending Rs 9017 million.

21. Correct option: (C) Carbon monoxide

The Ganga Action Plan or GAP was a program launched by Rajiv Gandhi in April 1986 to reduce the pollution load on the river. But the efforts to decrease the pollution level in the river became more after spending Rs 9017 million.

22. Correct option: (D) All

They are used in the manufacture of aerosol sprays, blowing agents for foams and packing materials, as solvents, ACs and as refrigerants in Refrigerators.

23. Correct option: (B) Acid Rain

Pollutants in the air, such as Sulphur dioxide and Nitrogen dioxide, react with water vapour in the atmosphere to produce Sulphuric acid and Nitric acid, respectively.

These acids react with rain, causing it to become acidic.

Marble cancer is a phenomenon caused by acid rain corroding building materials like marble.

24. Correct option: (A) Spring Water

High-quality spring water is rich in magnesium, potassium, calcium, sodium, and other trace minerals. It is naturally more alkaline than tap water, which can create an environment in the human body less vulnerable to disease proliferation.

25. Correct option: (C) Reduced oxygen carrying capacity of RBCs

This is because when carbon monoxide and oxygen both are available to RBCs then the RBCs will bind 200 times faster with carbon monoxide than oxygen, thus forming carboxyhaemoglobin rather than oxyhaemoglobin. This carboxyhaemoglobin reduces the oxygen carrying capacity of RBCs (since RBCs contain haemoglobin, a red-coloured protein).

26. Correct option: (B) By burning petrol in automobiles

Leaded petrol used to be a source of high levels of lead in the air of major cities. Significant reductions in the levels of airborne lead were achieved over recent years as the result of air pollution legislation introduced to remove lead from petrol.

27. Correct option: (A) Increasing Forest reserves

Burning fossil fuels like coal release carbon dioxide and cause a large amount of air pollution. The construction of the housing estates releases harmful chemicals which result in air pollution. The quarrying of the limestone also results in the release of the particles causing air pollution. The increase in the forest reserves does not result in any kind of pollution. Rather planting more trees helps to control air pollution.

28. Incorrect options given

29. Correct option: (D) iii, i, ii

The amount of water vapor in the air is known as humidity. During condensation of the water molecules, water vapor combines to make tiny liquid water droplets that hang in the air. As we know water vapor is a gas and invisible but because of the tiny water droplets, we can see the fog with our eyes. Asthma is a chronic lung condition caused by Smog. Smog contains Smoke + Fog + Fumes.

30. Correct option: (C) iii, i, iv, ii

Carbon dioxide is greenhouse gas. CO_2 is strongly absorbed by Infrared radiation. Nitrogen oxides affect vegetation. Sulphur dioxide is potent air pollutant.